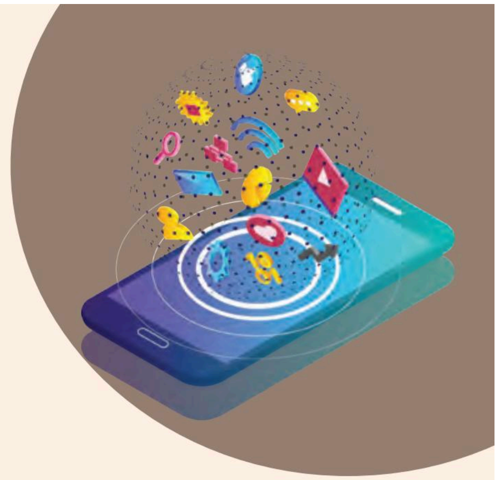


10

Creating a Spreadsheet

A spreadsheet or worksheet is a file used for storing, organising, and manipulating data in a tabular format. It is also called an electronic spreadsheet. A spreadsheet is used to perform calculations using formulas and functions. Spreadsheets also allow you to analyse the data visually with the help of charts. OpenOffice Calc is a spreadsheet program that interactively manages and organises data in rows and columns.

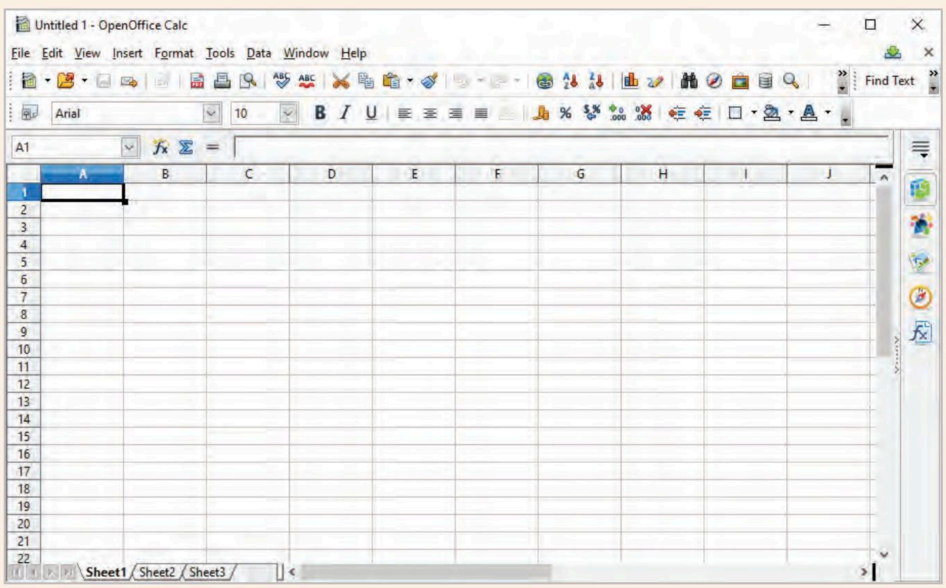


Fig. 10.1: A Spreadsheet

General Features of a Spreadsheet

There are a number of features that are available in a spreadsheet to perform tasks in an easy way. Some of the main features are:

1. **Grid Interface:** Spreadsheets are organised as a grid of rows and columns, where cells are used to input and manipulate data. Each cell is uniquely identified by a row and column label.

2. **Data Entry:** A spreadsheet allows users to enter various types of data, including numbers, text, dates, and formulas, into individual cells. This data can be manipulated, calculated, and formatted.
3. **Functions:** Functions are inbuilt formulas that are used for doing calculations upon the data entered. Commonly used functions are sum, average, max, min, count, etc.
4. **AutoFill:** The AutoFill feature allows you to quickly fill cells with repetitive or sequential data such as chronological dates, numbers, or repeated text. AutoFill can also be used to copy formulas.
5. **AutoSum:** This feature helps you to add up the numeric values in a group of selected cells.
6. **Charts:** Charts are used to represent the numerical data in a graphical manner. Charts help users analyse data in an easier way. Some commonly used charts are column, pie, bar graphs, etc.
7. **Data Formatting:** Spreadsheets allow users to format data in cells to control how it is displayed, including options for number formatting, font styles, alignment, borders, and fill colours.
8. **Data Validation:** Spreadsheets enable users to ensure data consistency and accuracy, helping users prevent errors during data entry.
9. **Import and Export Data:** Spreadsheets allow you to import or export data in multiple formats such as PDF, HTML, CSV, Postscript, etc.
10. **Data Protection:** Spreadsheets often offer password protection and access control to restrict who can view or edit certain parts of the document.

Did You Know?

Daniel Bricklin and Bob Frankston were the first Americans who discovered an electronic spreadsheet, which they named VisiCalc in early 1980s.

Popular Spreadsheet Software Examples

Many types of spreadsheet applications have been developed by various software corporations. Some of these applications are:

1. Microsoft Excel
2. LibreOffice Calc
3. OpenOffice Calc
4. Gnumeric
5. Google Sheets
6. Apple Inc. Numbers
7. wikiCalc

OpenOffice Calc is the spreadsheet application of the Apache OpenOffice suite. This is an open-source program that can be used on multiple operating systems, such as Windows, Linux, etc.

Starting the OpenOffice Calc Application

To start the OpenOffice Calc application, perform the following steps:

1. Click on the **Start** button. The Start menu appears.
2. Click on the **All apps** option in the Start menu. A list of installed apps is displayed.
3. Select the **OpenOffice Calc** option from the list of apps.

The **OpenOffice Calc** application window displays the three worksheets by default: Sheet1, Sheet2, and Sheet3.

You can also open the OpenOffice Calc application window by typing Calc or OpenOffice Calc in the search bar and then clicking on the OpenOffice Calc icon.

Components of an OpenOffice Calc Spreadsheet

The OpenOffice Calc application window comprises the following components:

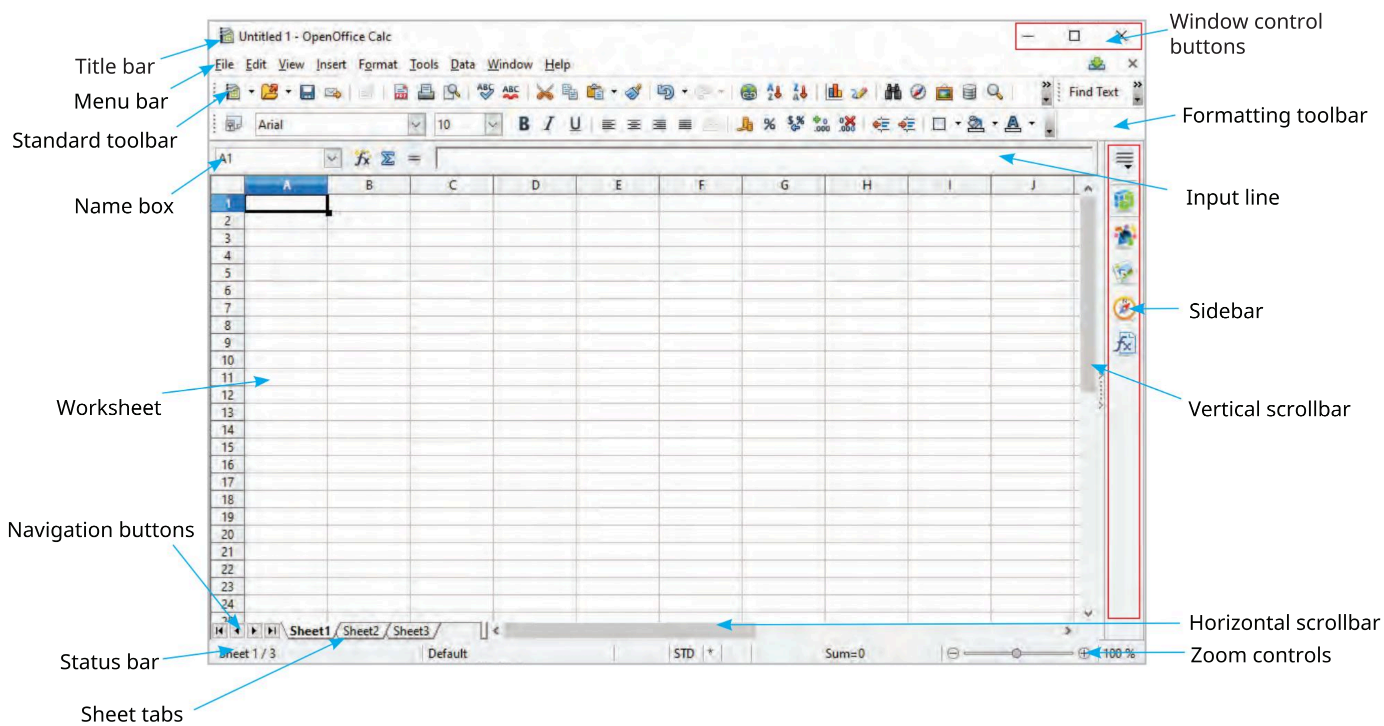


Fig. 10.2: Components of a Spreadsheet

Let us learn about each of these components:

1. **Title Bar:** At the top of a worksheet, there is a title bar that bears the name of the worksheet. By default, the name of the worksheet is displayed as **Untitled 1**. The name can be changed by the user.
2. **Window Control Buttons:** These buttons are used to minimize, maximize/restore and close the OpenOffice Calc window.
3. **Workbook and Worksheet:** A workbook can be imagined as a notebook with multiple pages in it. The pages can be said to be similar to worksheets. A worksheet is a grid of cells arranged in rows and columns. Data can be entered in each of these cells. By default, a workbook contains three worksheets. We can add more worksheets to a workbook.

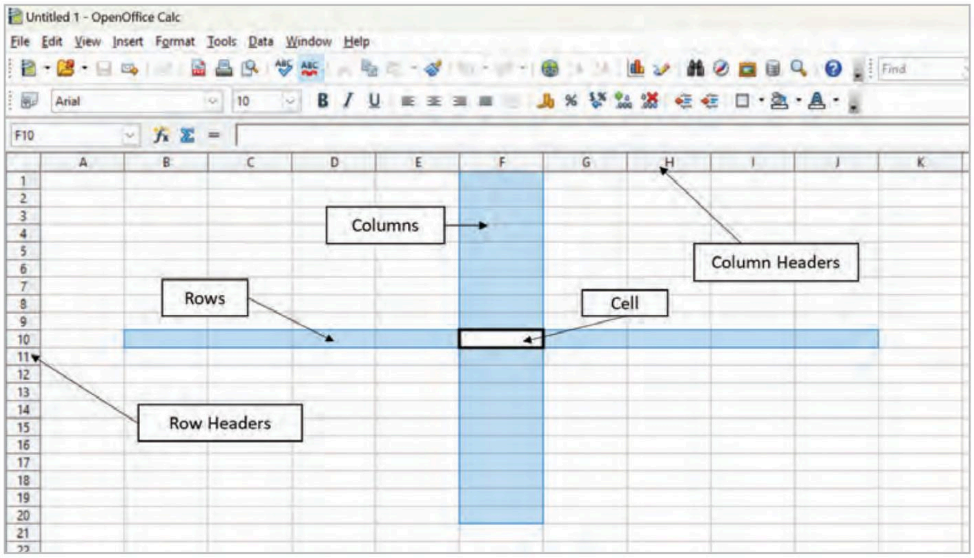


Fig. 10.3: Rows and Columns in a Calc Window

- **Active Cell:** When a cell is clicked, it gets highlighted by a thick black border. This is called an active cell. Data can be entered into this cell only. In Fig. 10.3, the cell in column F and row 10, called cell F10, is the active cell.
- **Column Headings:** Every column is marked by a head cell at the top. These head cells are marked from A to AMJ. There are a total of 1024 columns in a worksheet.
- **Row Headings:** Every row has a head cell at the extreme left. These are marked as numbers. There are 10,48,576 rows in a worksheet.
- **Cell:** The grid is formed by continuous cells running in vertical columns and horizontal rows. Each cell in a worksheet has a unique cell address. For example, a cell with address E5 refers to the cell found at the intersection of column E and row 5. All data is entered in a cell.

4. **Menu Bar:** The menu bar contains a number of menus, which can be used to give commands to the spreadsheet software. Typically, a menu bar contains the following menus:

- **File:** This menu contains file-related commands that apply to the entire worksheet. Commonly used commands in the File menu are: New, Open, Save As, Export as PDF, Digital Signatures, Page Preview, Print, etc.

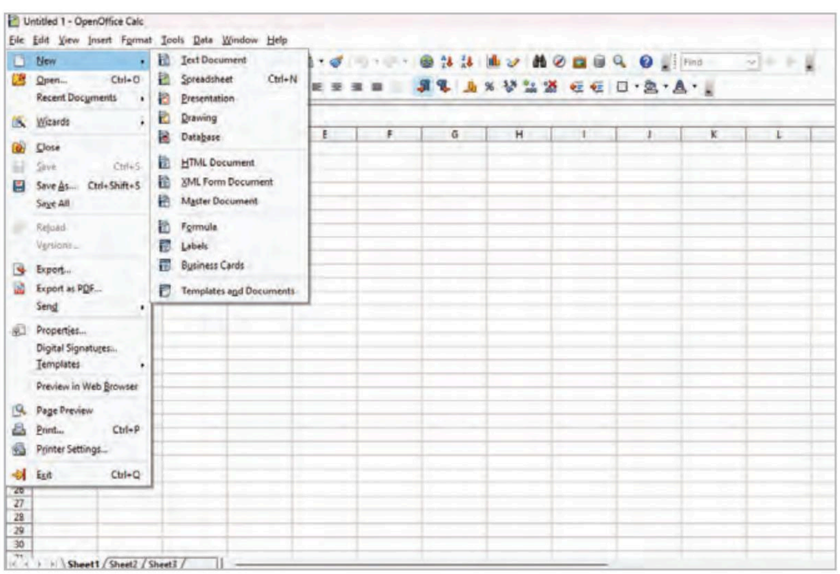


Fig. 10.4: The File Menu

- **Edit:** This menu contains the commands that are used for editing the content in the spreadsheet. Commonly used commands in the Edit menu are Cut, Copy, Paste, Find & Replace, Delete Cells, etc.

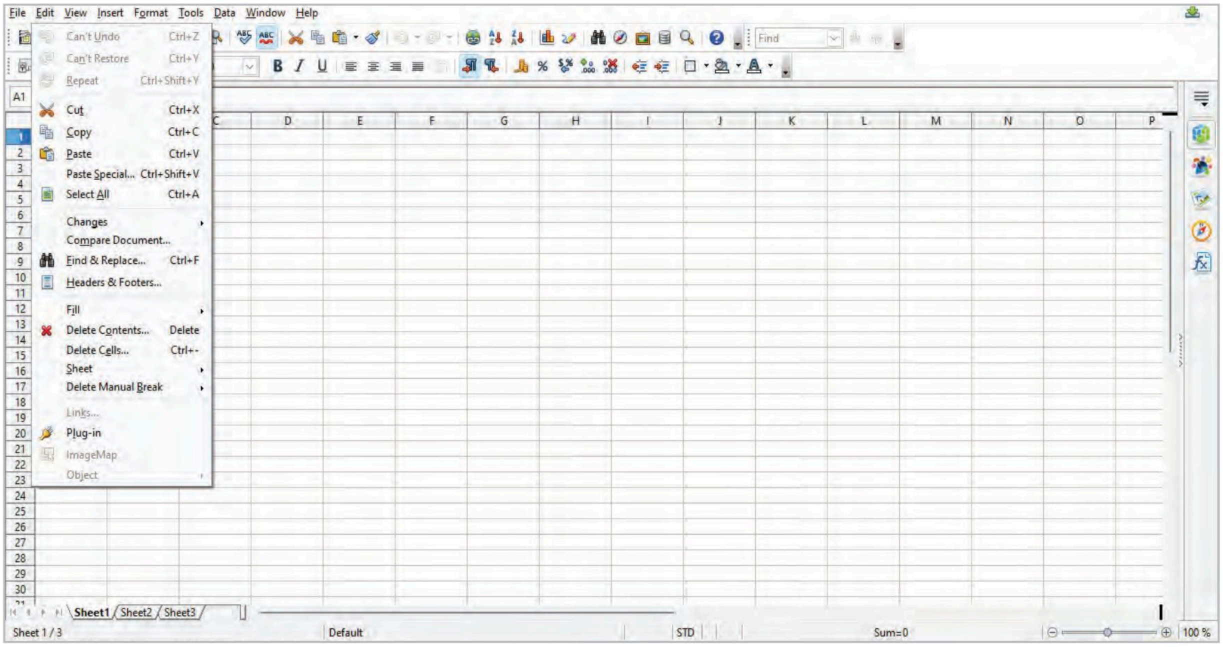


Fig. 10.5: The Edit Menu

- **View:** This menu contains the commands that can change the look of the spreadsheet. These commands are used for controlling the way you view and interact with your spreadsheet. Commonly used commands in the View menu are Page Break Preview, Full Screen, Zoom, etc.

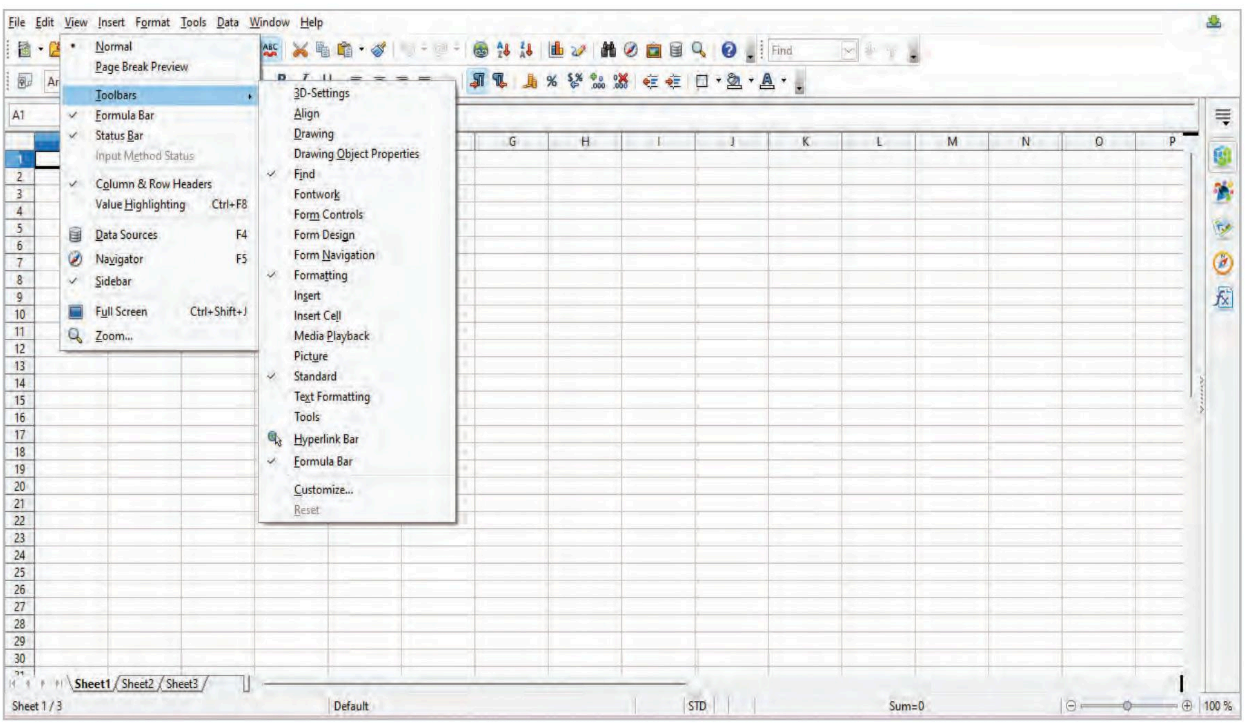


Fig. 10.6: The View Menu

- **Insert:** This menu contains the commands that allow you to add various elements and objects to your spreadsheet to enhance its functionality and appearance. You can insert **Cells**, **Rows**, **Columns**, **Pictures**, **Movies and Sounds**, **Charts**, etc., using the options under this menu.

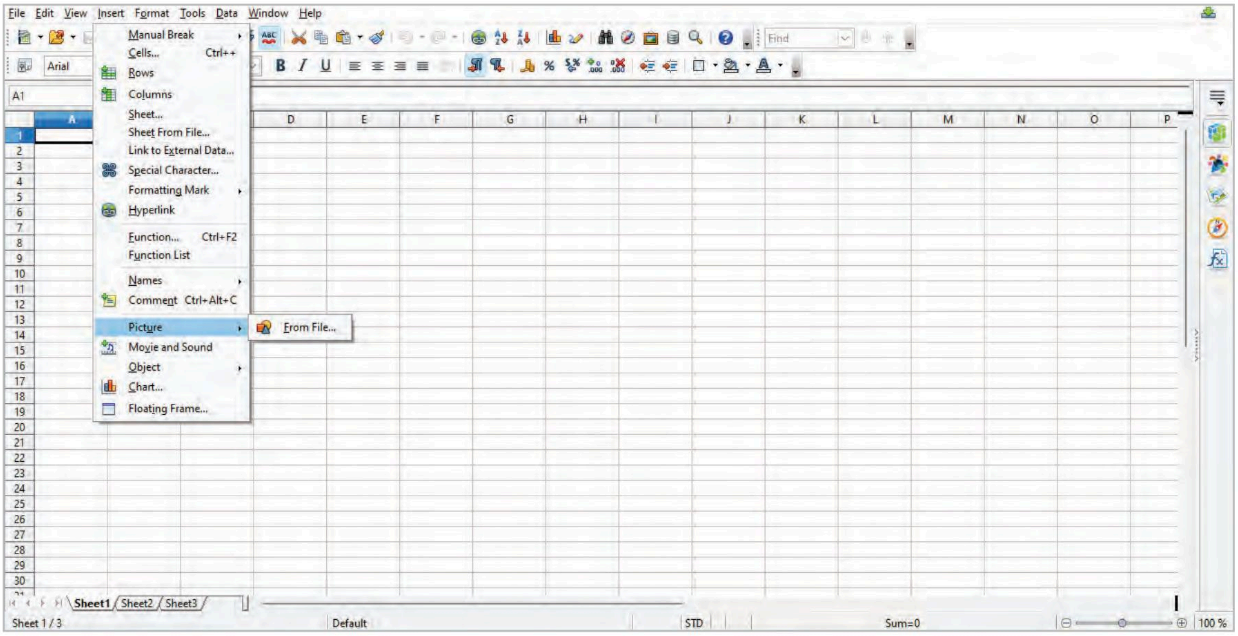


Fig. 10.7: The Insert Menu

- **Format:** The layout of the entire worksheet can be changed by using the **Format** menu. This menu contains commands for formatting and styling cells, rows, columns, and the overall appearance of the spreadsheet. Desired changes to the worksheet can be done with respect to **Cells**, **Merge Cells**, **Change Case**, **Styles and Formatting**, etc.

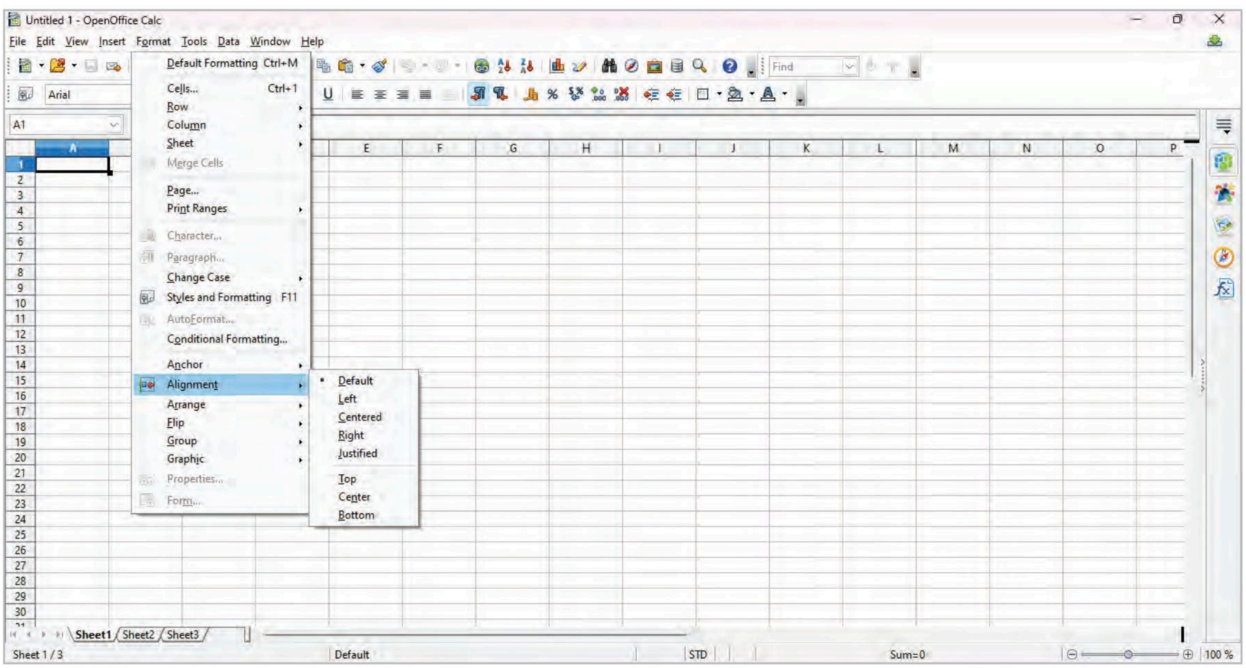


Fig. 10.8: The Format Menu

- **Tools:** Commands such as **Spelling**, **Goal Seek**, **Share Document**, **Merge Document**, **Cell Contents**, **Gallery**, **Macros**, etc., can be used by this menu for spreadsheet management, data analysis, and other functionalities.

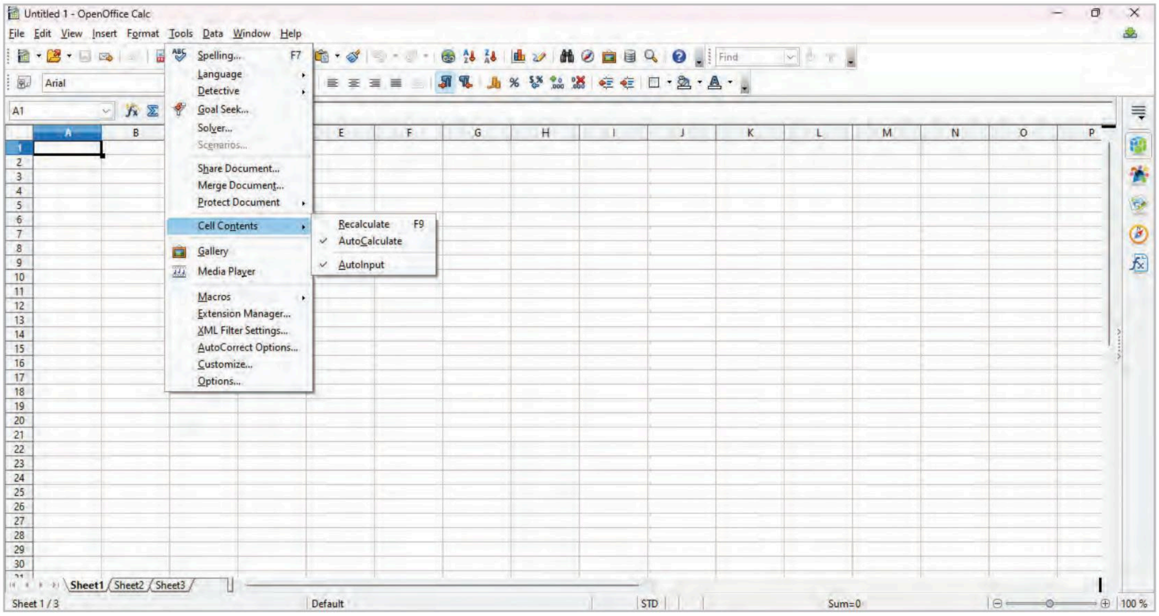


Fig. 10.9: The Tools Menu

- **Data:** This menu contains commands for the manipulation of data. Commonly used commands in this menu are **Define Range**, **Sort**, **Filter**, **Subtotals**, **Consolidate**, **Pivot Table**, etc.

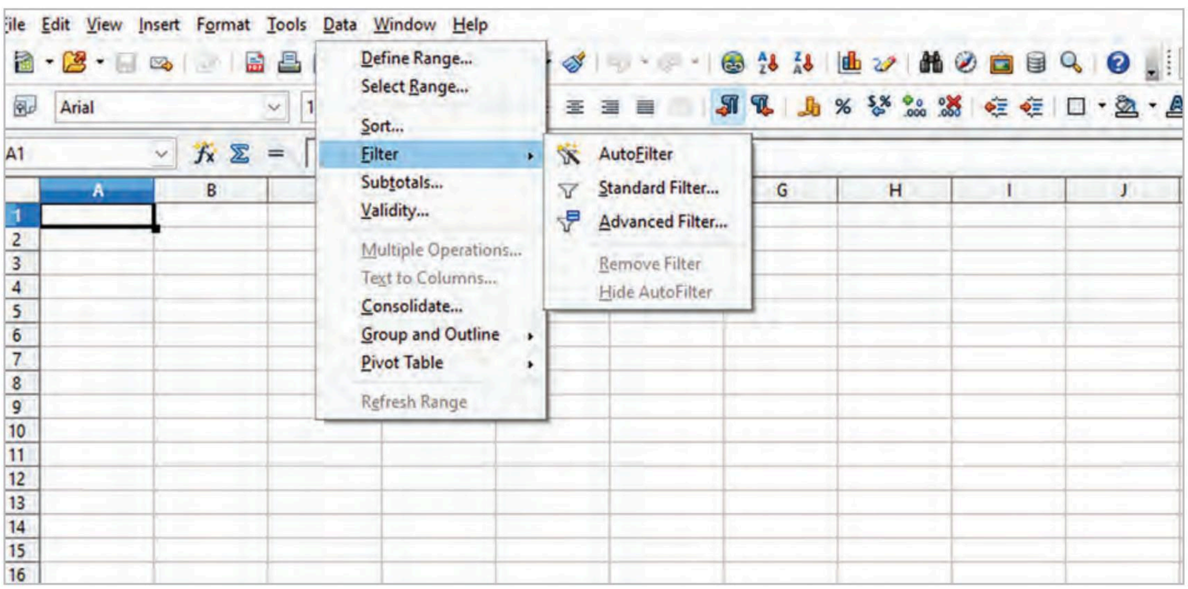


Fig. 10.10: The Data Menu

- **Window:** This menu contains commands for managing the windows and views within the application.

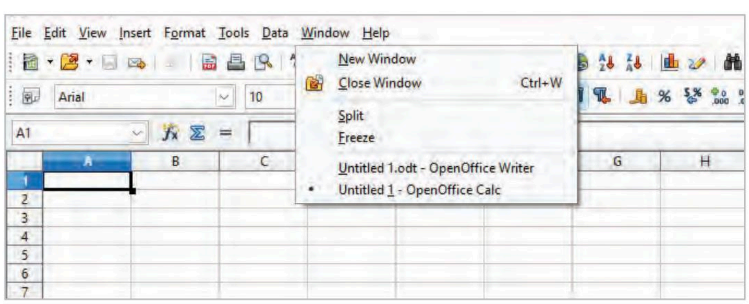


Fig. 10.11: The Window Menu

- **Help:** This menu contains commands that can assist users to get help, support, and information about the software.

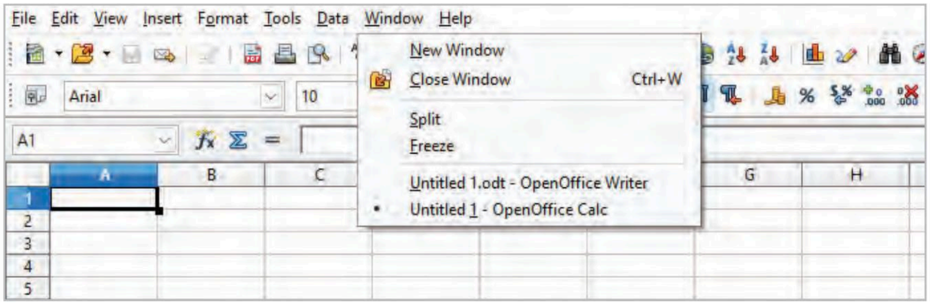


Fig. 10.12: The Help Menu

5. **Standard Toolbar:** A standard toolbar contains tools that are used for performing basic functions such as opening a new workbook, saving a workbook, printing a worksheet, inserting charts, sorting data, cutting, copying, pasting the selected data, etc.
6. **Formatting Toolbar:** This toolbar contains commands to format data. Some of the options that this toolbar provides are **Font Name**, **Font Size**, **Bold**, **Italic**, **Underline**, **Background Color**, **Font Color**, etc.
7. **Name Box:** The Name Box is present in the top left-hand corner of the worksheet. It displays the name or address of the active cell or range of cells.
8. **Input Line:** It displays the contents of the selected cell (data, formula, or function), thereby allowing editing of the cell contents. It is also known as the formula bar. To edit content using the Input line, just click in the Input line, and type in the changes. To edit within the current active cell, just double-click on the active cell and type in the changes.
9. **Function Wizard:** It helps to search for the desired function from the list of available functions.
10. **Sum:** This button is used to calculate the sum total of the numbers in the cells above the selected cell. The sum is entered in the active cell.
11. **Function:** Clicking on the Function icon inserts an equal to sign (=) in the active cell, and the Input line allows the formula to be entered.
12. **Status Bar:** The ribbon present at the bottom of worksheets that shows the current information of the worksheet is the Status bar. The status depicted by the status bar includes the sheet number of the current worksheet, the sum of the selected numeric cells, etc.
13. **Vertical Scrollbar:** The vertical scrollbar enables you to scroll the worksheet vertically, allowing you to move it up or down.
14. **Horizontal Scrollbar:** The horizontal scrollbar enables you to scroll the worksheet horizontally, allowing you to move it left and right.
15. **Navigation Buttons:** The navigation buttons allow you to navigate between different worksheets within a workbook.
16. **Zoom Controls:** This offers the capability to adjust the zoom level of the Calc worksheet, enabling you to zoom in for a closer view or zoom out for a broader perspective.
17. **Sidebar:** The sidebar displays icons for accessing various panels, such as **Properties**, **Styles and Formatting**, **Gallery**, **Navigator**, and **Functions**. These panels offer additional functionality and options for working with your spreadsheet.

Some useful key combinations to navigate within the same worksheet are:

Key or key Combination	Output
Arrow Keys	Move a single cell up, down, left, or right
Ctrl + Arrow Keys	Move the cell pointer to the first and last cell of a cell range in the direction of the arrow
Home	Moves to column A along the row where the active cell is selected
Ctrl + Home	Moves the cell to A1 position
Ctrl + End	Moves to the bottom-right cell of the data range
Page Up	Moves the worksheet one screen up
Page Down	Moves the worksheet one screen down

Range of Cells

When a number of adjacent cells are selected or denoted collectively in an area of a worksheet, this is called a range of cells. A range can be defined in a worksheet by writing the address of the starting cell, followed by that of the ending cell, separated by a colon (:). A cell range can be of three types:

1. **Column Range:** When the selected range of cells belong to a column, this is called a column range. When defining the range of such cells, the column letter remains the same, while the row number changes. For example, a column range that includes cells from rows 1 to 8 in column B can be written as B1:B8.

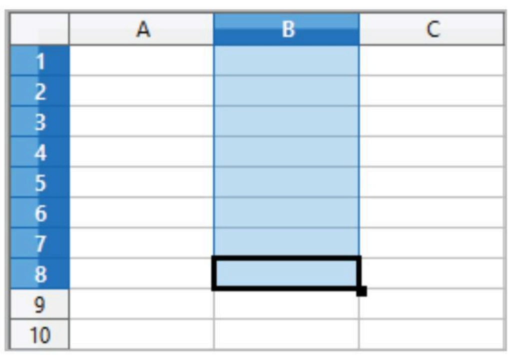


Fig. 10.13: Column Range

2. **Row Range:** The range of cells spread across a row is called the row range. In the row range, the row number remains the same while the letter of the column changes. For example, a range of cells that includes columns A to E of row 4 can be written as A4:E4.

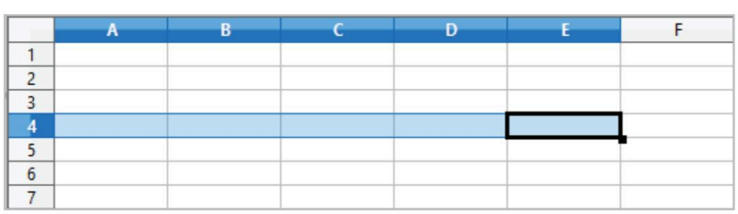


Fig. 10.14: Row Range

3. **Row and Column Range:** Such a range includes cells from multiple rows of multiple columns, so as to form a matrix. For example, a range of cells consisting of rows 1 to 9 of columns B and C can be written as B1:C9.

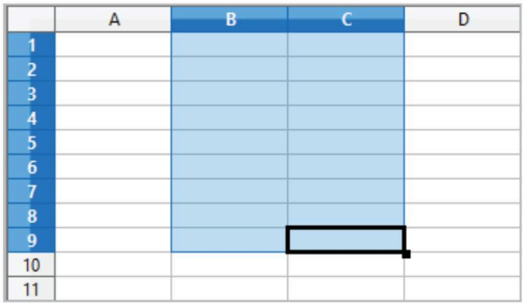


Fig. 10.15: Row and Column Range

Think and Tell

What are the advantages of using rows and columns to structure data in a spreadsheet?

Inserting a Worksheet

To insert a new worksheet, follow the steps:

1. Click the **Insert** menu and select the **Sheet** option from the context menu.

   OR Right-click on the Sheet tab and select the **Insert Sheet** option.

   OR click the blank area to the right of the Sheet tab.

   The **Insert Sheet** dialog box appears.

2. Select the desired position where you want to place the new worksheet.
3. Choose the number of sheets you want to insert.
4. Enter the name of the worksheet in the **Name** textbox.
5. Click the **OK** button.

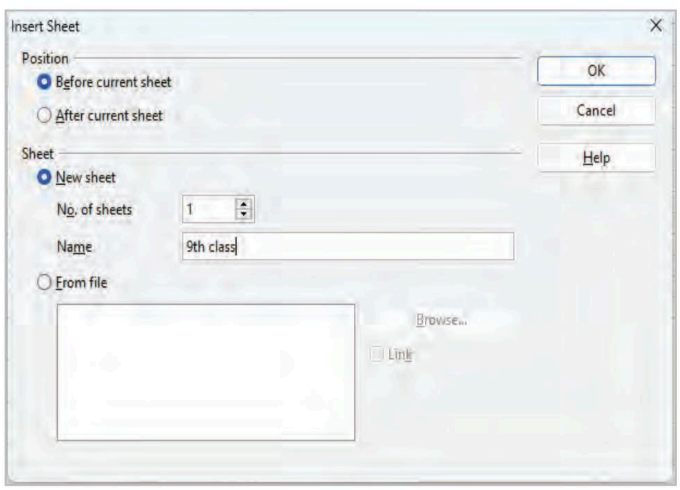


Fig. 10.16: Inserting a New Worksheet

Renaming a Worksheet

The steps to rename a spreadsheet are as follows:

1. Select the worksheet that you want to rename by clicking on the Sheet tab.
2. Click the **Format** menu and select the **Sheet** > **Rename** option from the context menu.

   OR Right-click on the space next to the Sheet tab and select the **Rename Sheet** option.

   The **Rename Sheet** dialog box opens.

3. Specify the name of the worksheet in the **Name** textbox.
4. Click the **OK** button.

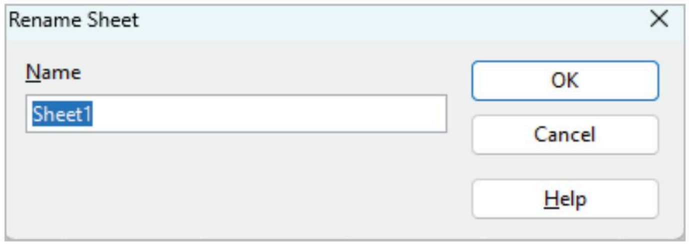


Fig. 10.17: Renaming a Worksheet

Now, the name of the worksheet gets changed.

Saving a Workbook

A workbook can be saved at the desired location by performing the following steps:

1. Click on the **File** menu and select the **Save** option.

   OR click on the **Save** button on the Standard toolbar.

   OR use the **Ctrl+S** shortcut keys.

   The **Save As** dialog box opens.
2. Select the desired location where you want to save the workbook.
3. Enter the desired name of the workbook in the **File name** text box.
4. Click the **Save** button.

The workbook is saved as a file with the extension .ods.

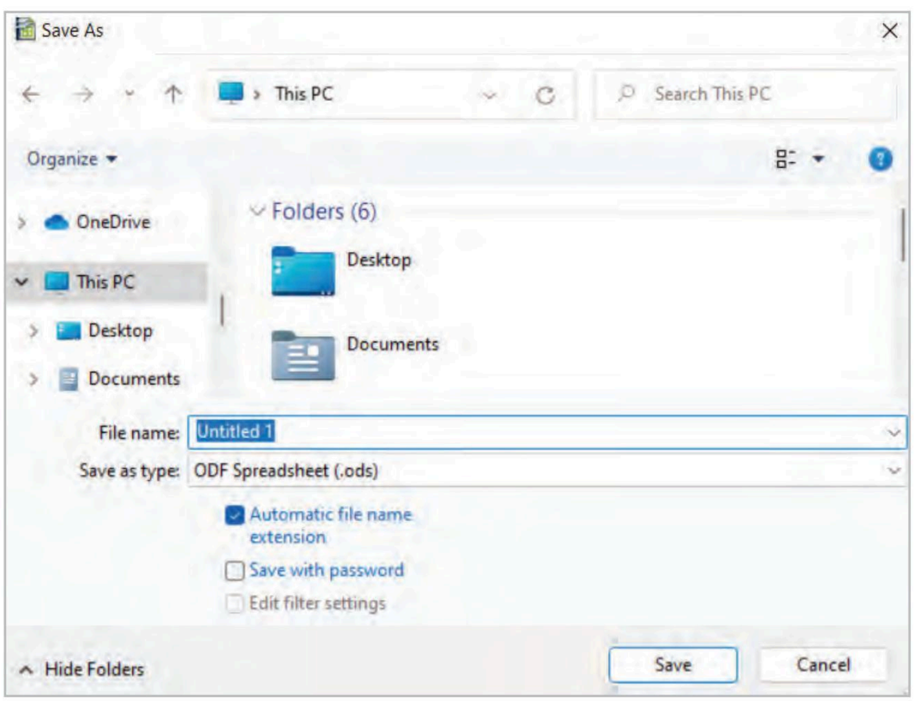


Fig. 10.18: Saving a Workbook

Deleting a Worksheet

OpenOffice Calc allows you to delete a worksheet in a workbook. However, if there is only one worksheet in a workbook, then the whole workbook has to be deleted. A worksheet can be deleted by performing the following steps:

1. Right-click on the Sheet tab of the worksheet that you want to delete and select the **Delete Sheet** option.
   A dialog box opens, asking for confirmation to delete.
2. Click **Yes** if you really want to delete the worksheet; else click **No.**

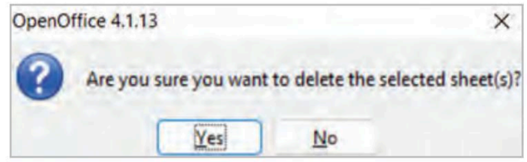


Fig. 10.19: Deleting a Worksheet

Closing a Workbook

A workbook can be closed by adopting any of the following methods:

Select the **File** menu and click **Close.**

OR

Click the **Close** × button on the title bar.

Activity Time

Activity 1: Group Discussion ***(Group Activity)***

In a group of 4-5 students, discuss how the Calc spreadsheet software can be harnessed for various purposes across different professional domains.

Activity 2: Conduct a Quiz Game ***(Group Activity)***

Conduct a collaborative quiz game where groups of two take turns asking each other to identify specific components of a Calc worksheet. Assign 10 points for each correct identification and deduct 5 points for every incorrect answer.

Activity 3: Create and Save Your Worksheet ***(Individual Activity)***

Open Calc spreadsheet software and perform the following steps:

- Note down the default number of worksheets in your workbook and insert one worksheet before the current worksheet.
- Rename the newly created worksheet with your name.
- Now, delete the first worksheet.
- Save the workbook with the name 'My_Workbook'. Observe the file extension with which it is saved.
- Close the workbook.

Chapter Checkup

A Select the correct option.

1. What is the primary purpose of a spreadsheet software like OpenOffice Calc?
   a. Word processing b. Data analysis and manipulation
   c. Graphic design d. Web browsing
2. Which feature of Calc allows you to visually represent numerical data?
   a. Data Entry b. Functions c. Charts d. AutoSum

3 What feature allows you to quickly copy sequential data in a spreadsheet?

a AutoSum
b AutoFill
c Data Validation
d Data Protection

B **Fill in the blanks with the most suitable words.**

1 In a .... range, the column letter remains the same while the row number changes.

2 The name or address of the active cell or range of cells is displayed in the ....

3 The default extension for a saved workbook in OpenOffice Calc is ....

4 The .... button is used to take the sum total of the numbers in the cells above the selected cell.

C **State whether the following is *True* or *False*. Correct the statements that are false.**

1 Spreadsheets are used for analysing data and performing calculations.

2 The Ctrl + Home key combination moves the cell to A1 position.

3 Data entry in spreadsheets can only include numbers and dates.

4 Spreadsheets can import and export data in only one format, such as PDF.

D **Answer the following questions. *(Solved)***

Q1. What is an active cell?

A1. When a cell is clicked, it gets highlighted by a thick black border. This cell is an active cell. Data can be entered into this cell only.

Q2. Explain the following terms:

a. Formatting toolbar

b. Name Box

c. Input line

A2. **a.** **Formatting Toolbar:** This toolbar contains commands to format data. Some of the options that this toolbar provides are Font Name, Font Size, Bold, Italic, Underline, Background Color, Font Color, etc.

b. **Name Box:** The Name Box is present on the top left-hand corner of the worksheet. It displays the name or address of the active cell or range of cells.

c. **Input Line:** It displays the contents of the selected cell (data, formula, or function), thereby allowing editing of the cell contents. It is also known as the formula bar.

Q3. Prabhav wants to create a calendar in a spreadsheet and he has to write down the names of the months along with the days of the week for the entire year. What feature should he use to prevent manual data entries in the spreadsheet?

A3. AutoFill feature

Answer Key

A **1.** b **2.** c **3.** b

B **1.** column **2.** Name Box **3.** .ods **4.** Sum

C **1.** True.

2. True.

3. False. Spreadsheets allow you to enter various types of data, including numbers, text, dates, and formulas.

4. False. Spreadsheets have the ability to import and export data in various formats, including but not limited to PDF, HTML, CSV, Postscript, and more.

11

Formulas and Functions in Spreadsheet

In this chapter, you will learn how to enter formulas and functions and perform various mathematical calculations. You should be able to create new charts and graphics using the input data. You can specify mathematical relationship(s) between the numbers using various formulas. Formulas are used for simple addition, subtraction, multiplication, and division, as well as for complex calculations. Functions are incorporated into the formula used for spreadsheets. The users need to provide cell references and addresses only. These are called arguments of the function and are given as enclosed between parentheses.

Spreadsheet Terminologies

To work upon the data and extract the desired useful information from a worksheet, some formulas and functions are applied on the data. In spreadsheets, formulas are used to calculate results from the input worksheet data. If there is some change or editing in the previously entered data, such formulas automatically recompute the updated results.

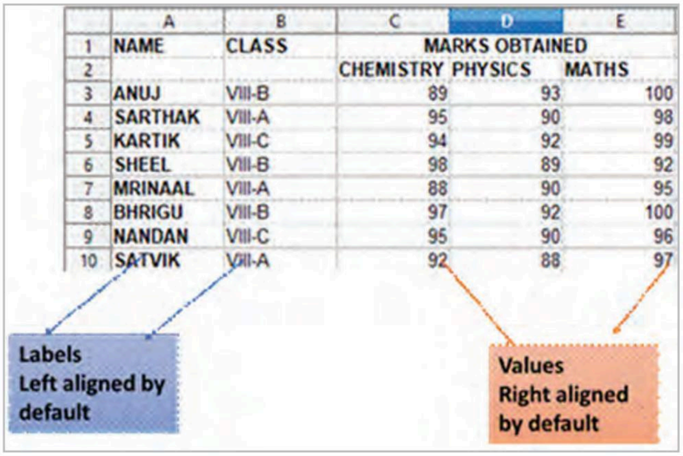
In order to get a complete understanding of formulas and functions, we have to understand some of the basic terminologies and features of the spreadsheet, such as:

Labels

A text or special character, descriptive information (non-numeric data), or a combination of numeric and non-numeric data is treated as a label. Labels cannot be treated by any mathematical operation, such as multiplication, subtraction, etc. Labels include any cell contents beginning with alphabets. Labels entered into a cell are left-aligned by default.

Value

The data, typically consisting of numeric inputs, is called value. Spreadsheets support performing calculations of various forms of data, such as integers, decimals, fractions, percentage data, etc. The numeric data entered into a cell is right-aligned by default.



	A	B	C	D	E
1	NAME	CLASS	MARKS OBTAINED		
2			CHEMISTRY	PHYSICS	MATHS
3	ANUJ	VIII-B	89	93	100
4	SARTHAK	VIII-A	95	90	98
5	KARTIK	VIII-C	94	92	99
6	SHEEL	VIII-B	98	89	92
7	MRINAAL	VIII-A	88	90	95
8	BHRIGU	VIII-B	97	92	100
9	NANDAN	VIII-C	95	90	96
10	SATVIK	VIII-A	92	88	97

Fig. 11.1: Labels and Values in a Spreadsheet

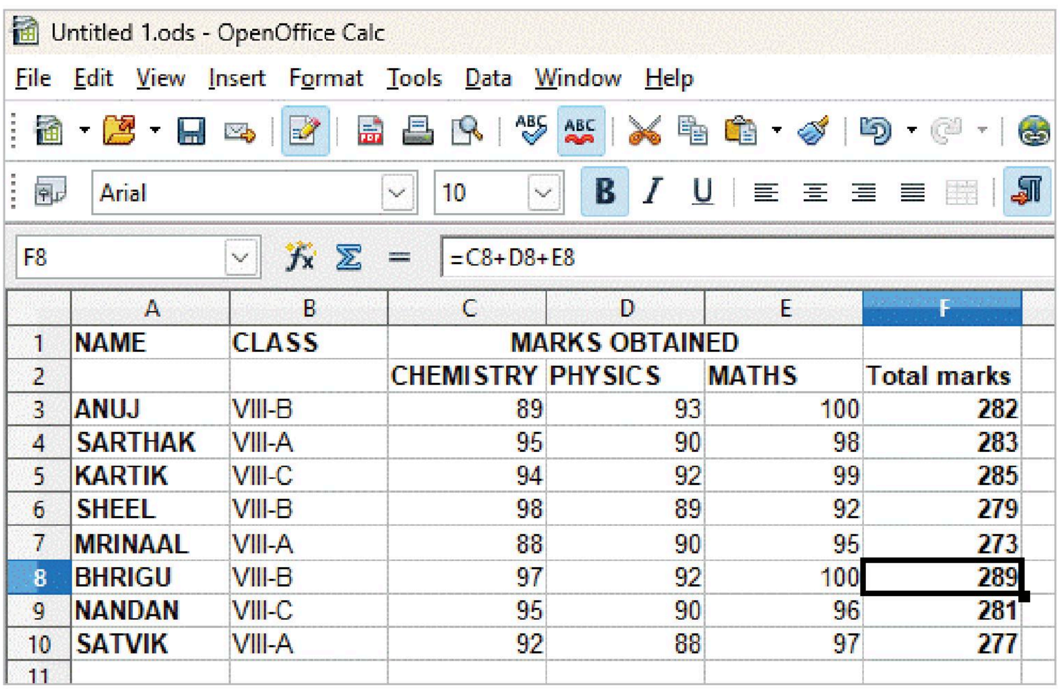
Did You Know?

The values do not display the preceding zeroes, when the data is value. In order to show a preceding zero, data has to be entered as text by entering an apostrophe before the number, e.g, to enter 0765 in a cell as a value, write it as '0765. Since such a figure is treated by the sheet as a text input, it becomes left aligned by default.

The numeric data entered in the form of digits from 0 to 9 are treated as positive values. A figure with a negative value has to be specified by entering the same in parentheses or adding a sign of negative before the figure. For example, (219) or '-219' represents negative value.

Formula

A formula is like a mathematical equation that performs calculations, returns information, or manipulates the contents of cells. The expression for a formula starts with an equal to sign (=), which is followed by values, cell addresses, functions, etc. On applying a formula to a cell, the value of the equation is displayed in the cell, and the formula is displayed in the Input line. A typical depiction of a formula applied to the given data is shown in Fig. 11.2.



	A	B	C	D	E	F
1	NAME	CLASS	MARKS OBTAINED			
2			CHEMISTRY	PHYSICS	MATHS	Total marks
3	ANUJ	VIII-B	89	93	100	282
4	SARTHAK	VIII-A	95	90	98	283
5	KARTIK	VIII-C	94	92	99	285
6	SHEEL	VIII-B	98	89	92	279
7	MRINAAL	VIII-A	88	90	95	273
8	BHRIGU	VIII-B	97	92	100	289
9	NANDAN	VIII-C	95	90	96	281
10	SATVIK	VIII-A	92	88	97	277
11						

Fig. 11.2: Applying Formula in a Spreadsheet

Working with Formulas and Functions

Calc can perform various types of calculations on the data provided. It can make data undergo addition, subtraction, multiplication, etc.

Working with Formulas

An expression of a formula involves the addresses of multiple cells, which need to undergo the required calculations. Both simple as well as complex calculations can be performed on the data. We need to know the various elements of a formula.

Calc can perform calculations utilising formula of three types:

1. **Basic Formula:** A basic formula involves only one type of operator, such as addition or subtraction. Consider an example given in Fig. 11.3. We need to perform the basic calculation of adding the data provided in cells A2 to A6 to get the resultant sum in the A7 cell. For performing basic calculations, follow the given steps:
   - Click on cell A7 where you want to display the result and type the '=' sign.
   - Click on the A2 cell. Type the '+' sign after that.
   - Repeat step b for cells A3 to A6 or simply type = A2+A3+A4+A5+A6 in the Input line.
   - Press the **Enter** key. The result will appear in cell A7.

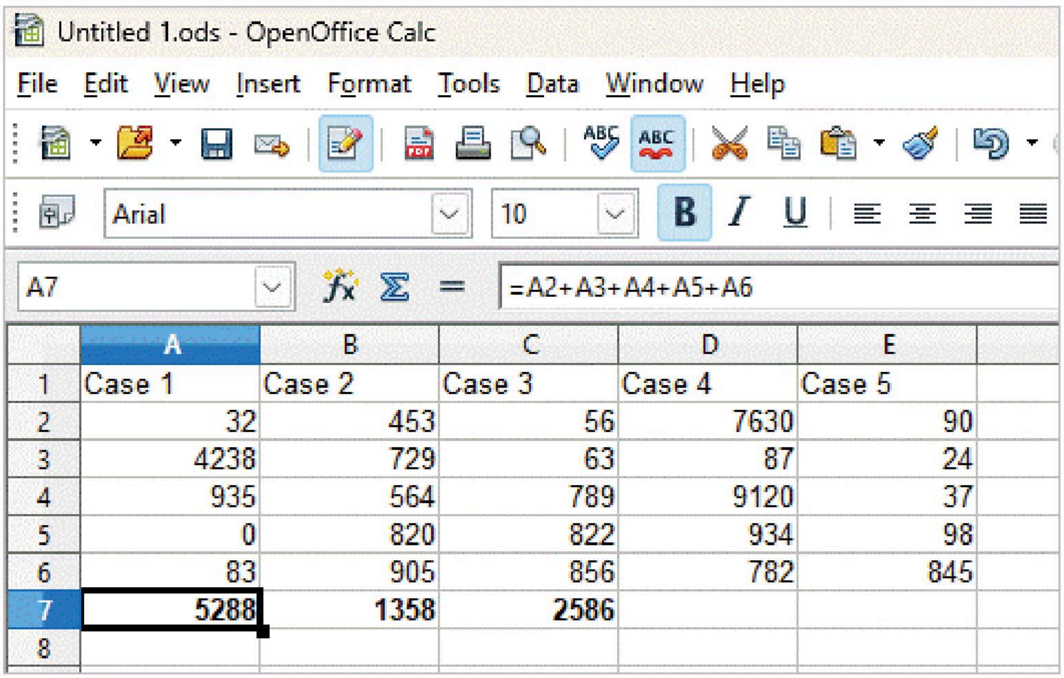
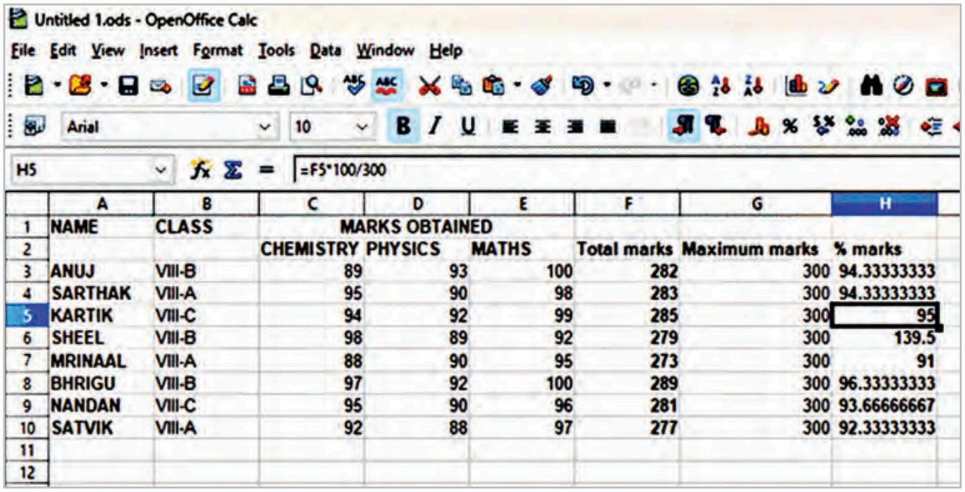


Fig. 11.3: Applying Basic Formula in Calc.

2. **Compound Formula:** Compound formulas use more than one operator in a single calculation. For example, refer to Fig. 11.4, where the marks obtained in the subjects of Chemistry, Physics and Maths of a certain number of students in class 8 are given. Column F contains the total marks obtained by these students, and we need to calculate the percentage of marks obtained by them in column H. Follow the given steps:
   - Click on H3 and type the '=' sign.
   - Type the formula: =F3*100/300.
   - Press the **Enter** key. The H3 cell displays the desired percentage of marks obtained by Anuj in the three subjects.
   - Repeat the steps as above for calculating the percentage of marks of individual students.

   Note that the result appears in the desired cell, and the formula is displayed in the Input line.



	A	B	C	D	E	F	G	H
1	NAME	CLASS	MARKS OBTAINED					
2			CHEMISTRY	PHYSICS	MATHS	Total marks	Maximum marks	% marks
3	ANUJ	VIII-B	89	93	100	282	300	94.33333333
4	SARTHAK	VIII-A	95	90	98	283	300	94.33333333
5	KARTIK	VIII-C	94	92	99	285	300	95
6	SHEEL	VIII-B	98	89	92	279	300	139.5
7	MRINAAL	VIII-A	88	90	95	273	300	91
8	BHRIGU	VIII-B	97	92	100	289	300	96.33333333
9	NANDAN	VIII-C	95	90	96	281	300	93.66666667
10	SATVIK	VIII-A	92	88	97	277	300	92.33333333
11								
12								

Fig. 11.4: Applying Compound Formula

3. **Text Formula:** We can also apply formulas on text or string data. The symbol ampersand (&) is used to join the text data. Joining two or more strings of text data is called **Concatenation.** For example, in the sheet shown, we need to join the text given in A2, B2, and C2. Follow the given steps:
   - Enter the data in cells A2, B2, and C2 as shown in Fig. 11.5.
   - Enter the formula '=A2 & B2 & C2' in cell D2 and press the **Enter** key.
   - D2 displays the text joined together as 'SwachhaBharatAbhiyaan'.

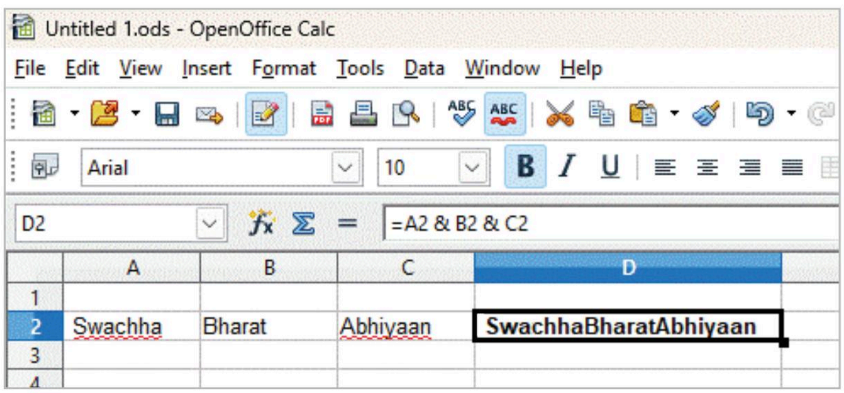


Fig. 11.5: Applying Text Formula

Elements of a Formula

A formula can be complex, which involves multiple functions applied to multiple cells. The structure of such a formula can be understood by knowing its constituents. For example:

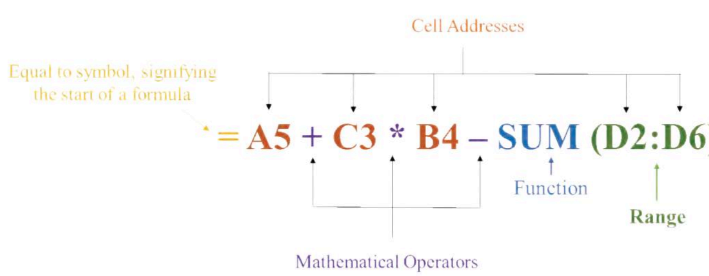


Fig. 11.6: Elements of a Formula in Calc.

References: A cell or a range of cells included in a formula for performing calculations

Mathematical Operators: Symbols of calculations such as '+', '-', '*', '/', and '^'

Constants: The values of the data or text which do not change

Functions: The operations which are predefined as formula in Calc.

Many operators and functions can be used in a formula as required.

Mathematical Operators	Operation
+	Addition
-	Subtraction
*	Multiplication
/	Division
^	Exponent (Power)

The formula is evaluated by the conventional precedence only. The conventional precedence is as shown in the following table:

Precedence	Operator
First	( )
Second	^
Third	/, *
Fourth	+, -

Consider the calculations performed in the following table:

Formula	Evaluation	Result
= 6 + 4 * 3	= 6 + 12	18
= (6 + 4) * 3	= 10 * 3	30
= 2*4^3	= 2*64	128
= (2*4)^3	= 8^3	512
= (9/3)^2	= 3^2	9
= 9/(3^2)	= 9/9	1

Did You Know?

Fill handle is a small square in shape, present on the bottom right edge of a cell. Fill handle can be used to drag and fill the adjacent cells with the same formula, value or data as entered in the selected cell.

Working with Functions

Functions are predesigned formula which are built into Calc for performing accurate calculations. A function is constituted by two parts: Argument and Structure.

Arguments: Arguments are the values input to the functions. These can be in the form of Text, Numbers, Logical Values (True or False), Constants, Range of Cells, or some other functions as well.

Structure: The structure of a function acts as its basic skeleton. On the basis of this structure, a general syntax can be written that represents the actual function. The structure of a function begins with the name of the function, followed by one or multiple arguments separated by a comma. When the structure is made of some range of cells, then the same are separated by semi-colons, presented within brackets. For example, the typical structure of a function can be written as:

=Function name (Argument1, Argument2, Argument3, ...)

Some commonly used functions are:

SUM( ): The SUM function is used to add the values of the selected cells or a range of cells. Follow the given steps to use this function:

1. Click on the cell where you want to get the result of the addition operation. Say C7 in the case given.
2. Type the '=' sign.
3. Type the name of the function, i.e., sum.
4. Type open parenthesis '('.
5. Type the range of cells and close the parenthesis, such as '=SUM(C2:C6)'.
6. Press the **Enter** key and see the result displayed in cell C7, as shown in Fig. 11.7.

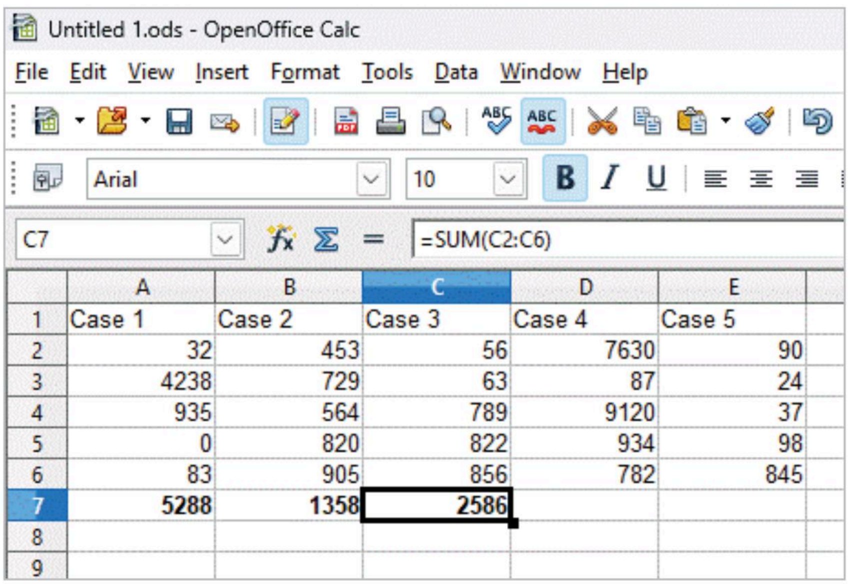


Fig. 11.7: SUM Function

Average ( ): The Average function is used to find the average or mean value of the selected cells or a range of cells. Follow the given steps to use the average function:

1. Click on the cell where you want to get the result of the average function. Say D7 in the case given (see Fig. 11.8).
2. Type the '=' sign.
3. Type the name of the function, i.e., average.
4. Type open parenthesis '('.
5. Type the range of cells and close the parenthesis, such as '=AVERAGE(D2:D6)'.
6. Press the **Enter** key and see the result displayed in cell D7, as shown in Fig. 11.8.

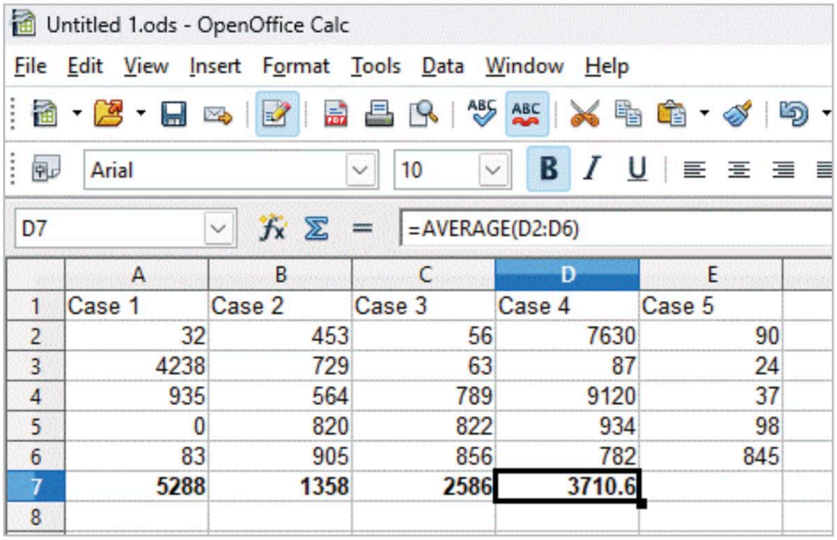


Fig. 11.8: AVERAGE Function

Count ( ): The Count function gives us the numeric count of the entered values of the selected cells. Follow the given steps to use the count function:

1. Click on the cell where you want to put the result for the count function. Say E7 in the case given (see Fig. 11.9).
2. Type the '=' sign.
3. Type the name of the function, i.e., count.
4. Type open parenthesis '('.
5. Type the range of cells and close the parenthesis, such as "=COUNT(E2:E6)".
6. Press the **Enter** key and see the result displayed in cell E7, as shown in Fig. 11.9.

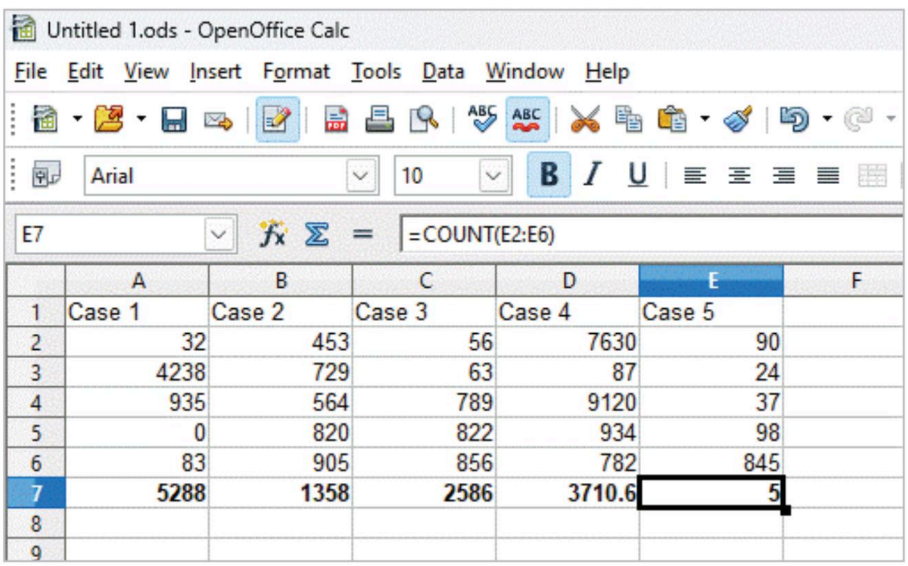


Fig. 11.9: COUNT Function

MAX ( ): This function gives us the largest of the entered values of the selected cells. Follow the given steps to use the max function:

1. Click on the cell where you want to put the answer to the max function. Say B8 in the case given (see Fig. 11.10).
2. Type the '=' sign.

3. Type the name of the function, i.e., max.
4. Type open parenthesis '('.
5. Type the range of cells and close the parenthesis, such as '=MAX(B2:B6)'.
6. Press the **Enter** key and see the result displayed in cell B8, as shown in Fig. 11.10.

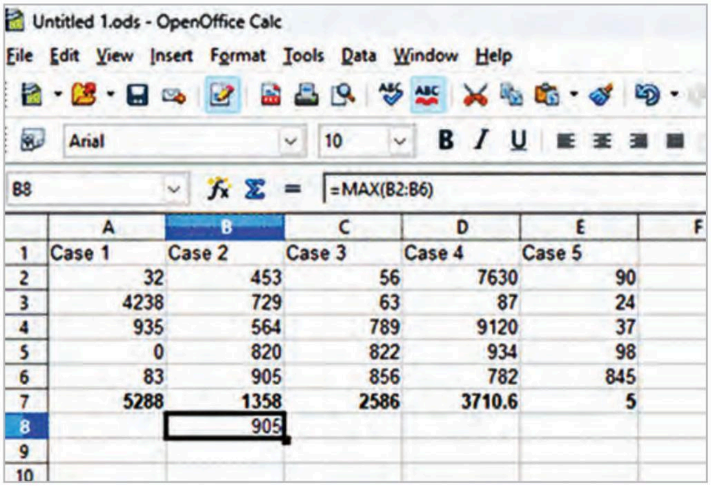


Fig. 11.10: Using the MAX Function

MIN ( ): This function gives us the smallest of the entered values of the selected cells. This function takes into consideration both the numeric entries as well as the logical values as arguments. If there are no valid arguments in the selected range of cells, the function returns a result of zero ['0']. Follow the given steps to use the min( ) function:

1. Click on the cell where you want to put the result for the min function. Say B9 in the case given (see Fig. 11.11).
2. Type the '=' sign.
3. Type the name of the function, i.e., min.
4. Type open parenthesis '('.
5. Type the range of cells and close the parenthesis, such as '=MIN(B2:B6)'.
6. Press the **Enter** key and see the result displayed in cell B9, as shown in Fig. 11.11.

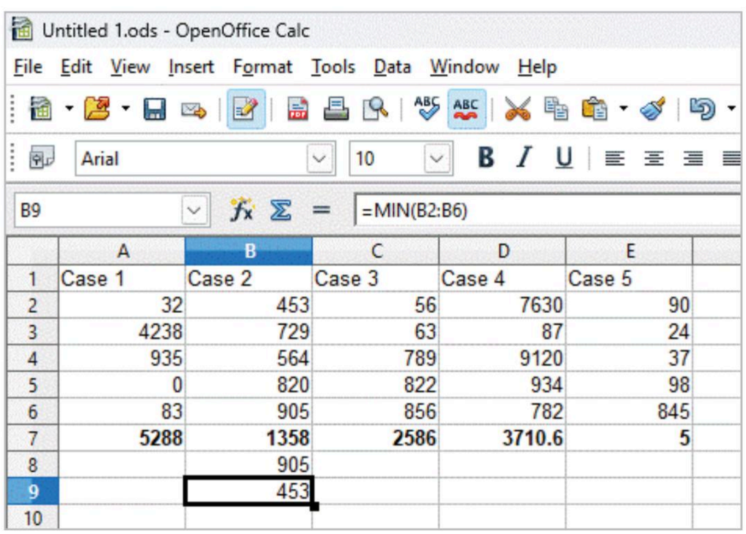


Fig. 11.11: Using the MIN Function

TODAY ( ): This function displays the current date. Follow the steps to use the Today function:

1. Click on the cell where you want to put the result for the Today function. Say H8 in the case given (see Fig. 11.12).
2. Type the '=' sign.
3. Type the name of the function, i.e., TODAY.
4. Type open parenthesis '('.
5. Type close parenthesis, such as '=TODAY()'.
6. Press the **Enter** key and see the result displayed as a date in cell H8, as shown in Fig. 11.12.

Following are the functions to display the current day, month, and year:

1. For day: DAY(TODAY())
2. For Month: MONTH(TODAY())
3. For Year: YEAR(TODAY())

In Fig 11.12, you can see that entering the formula '=MONTH(TODAY())' will display the current month.

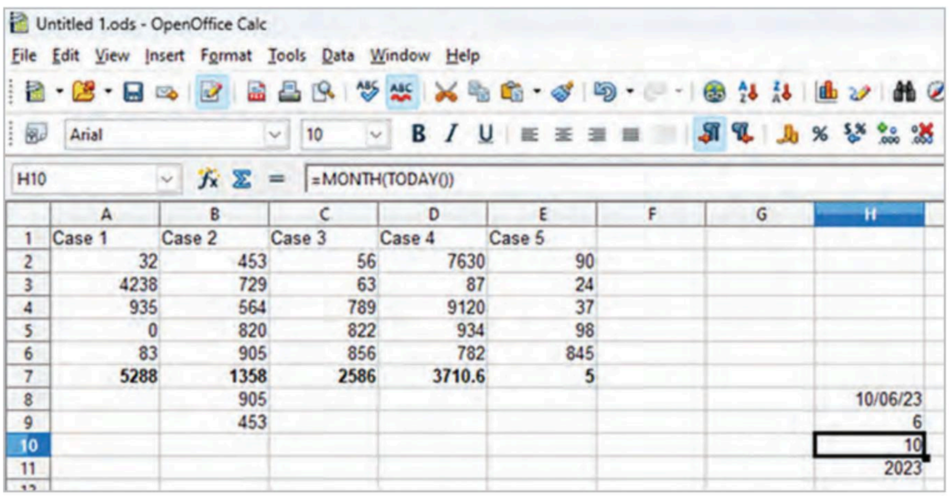


Fig. 11.12: Depiction of Today's Date Display in Calc

Date and time are entered into the active cell, separated by hyphens, slashes, or spaces. For example, 26th of January 2023 can be written as 26/Jan/2023, 26/01/2023, or 26-jan-2023. Calc recognises a variety of date formats. The elements of time can be separated from one another by using colons, such as 11:33:25.

Inserting Cells, Columns, and Rows

While entering data, one might need to add more cells, columns, or rows. Calc provides you with this facility.

Inserting New Cells

There are two methods to insert new blank cells in a spreadsheet.

1. Select the range of cells where you need to add more cells.
2. Right-click on the range of selected cells and choose the **Insert** option.

or alternatively,

1. Go to the **Insert** tab on the Menu bar.
2. Select the **Cells** option.

In both the cases, an **Insert Cells** dialog box appears (see Fig. 11.13), showing four options for the placement of new cells:

1. **Shift cells down:** This option moves the selected cell(s) down by one row (Fig. 11.14).
2. **Shift cells right:** This option moves the selected cell(s) right by one column (Fig. 11.15).
3. **Entire row:** Clicking this option lets you add rows to the entire sheet in the selected cell region (Fig. 11.16).
4. **Entire column:** Adds another column of new cells to the left of the selected cells (Fig. 11.17).

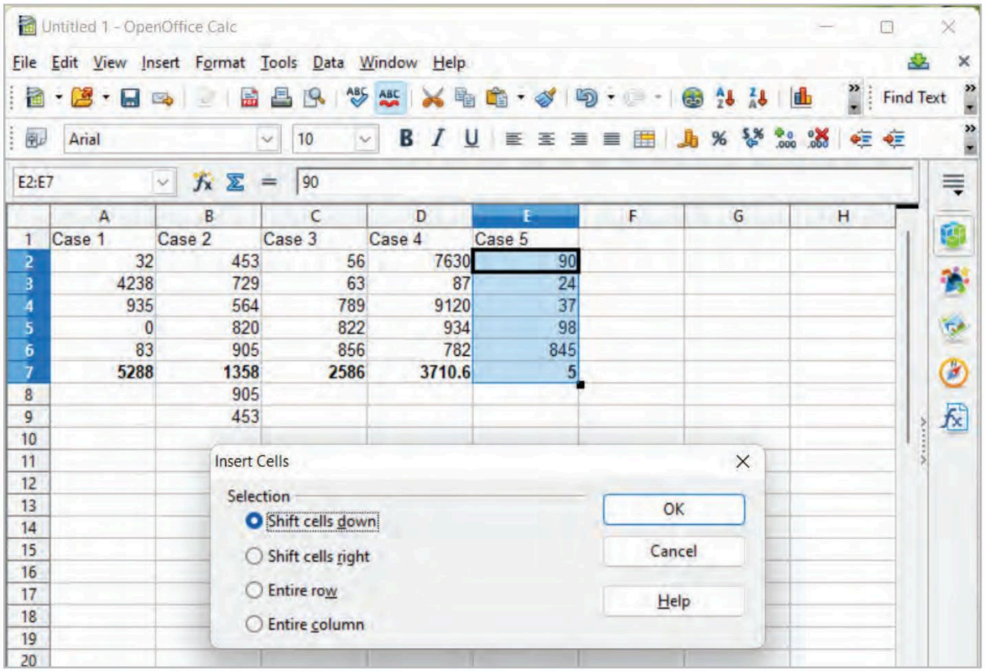


Fig. 11.13: Adding New Cells

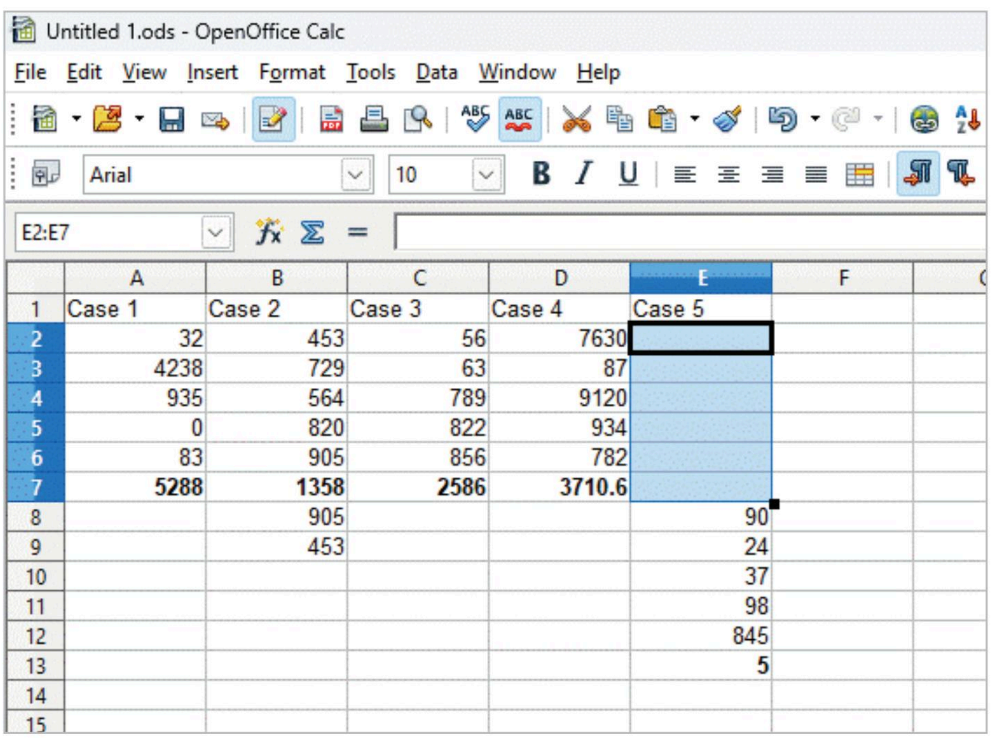


Fig. 11.14: Result of Selecting 'Shift cells down' Option

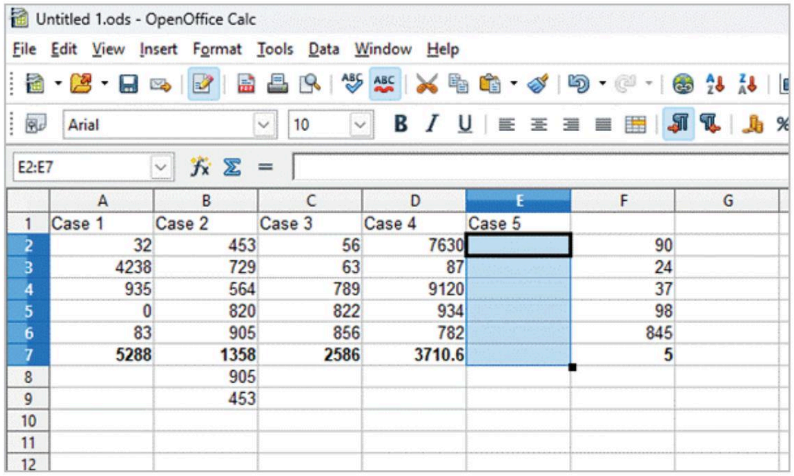


Fig. 11.15: Result of Selecting 'Shift cells right' Option

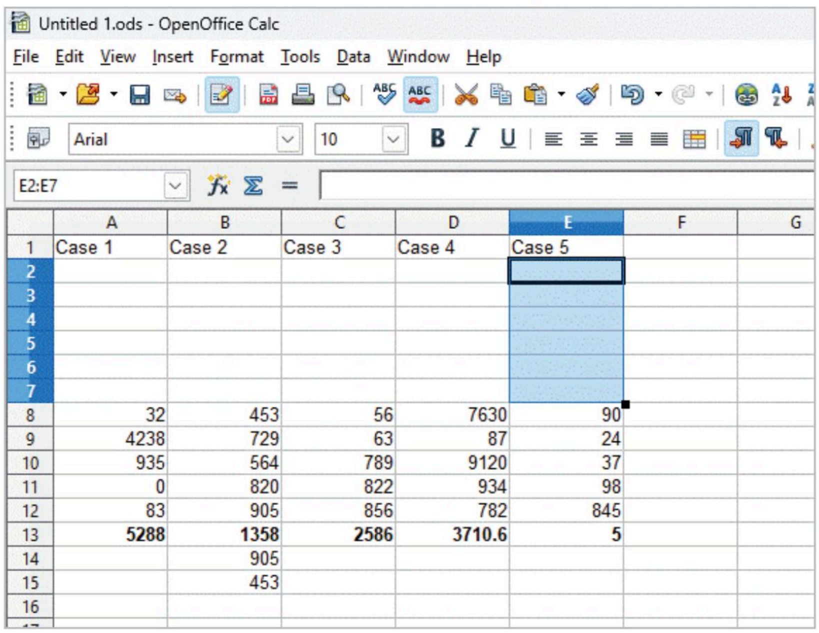


Fig. 11.16: Result of Selecting 'Entire row' Option

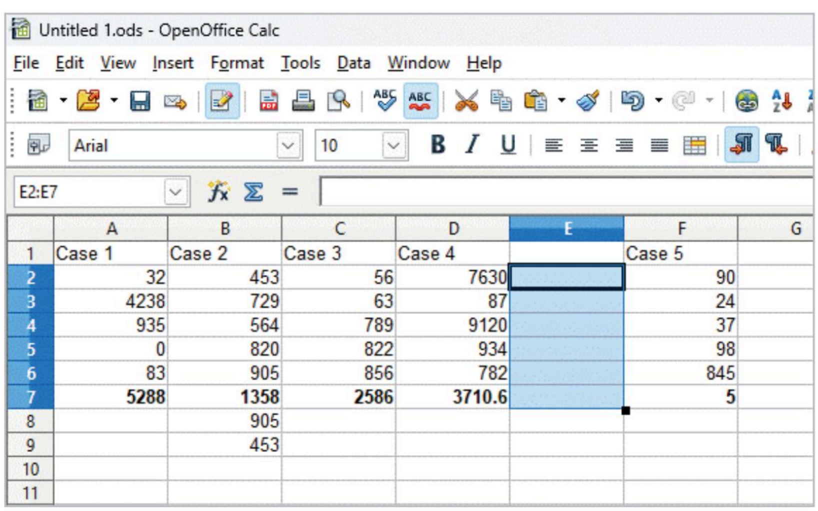


Fig. 11.17: Result of Selecting 'Entire column' Option

Inserting a Column

To insert a column, follow the steps given below:

1. Right-click on the column header where you need to add a new column (Column C, for example).
2. Select the **Insert Columns** option.

   OR

   click on the **Columns** option from the **Insert** menu.
3. A new column C gets added.

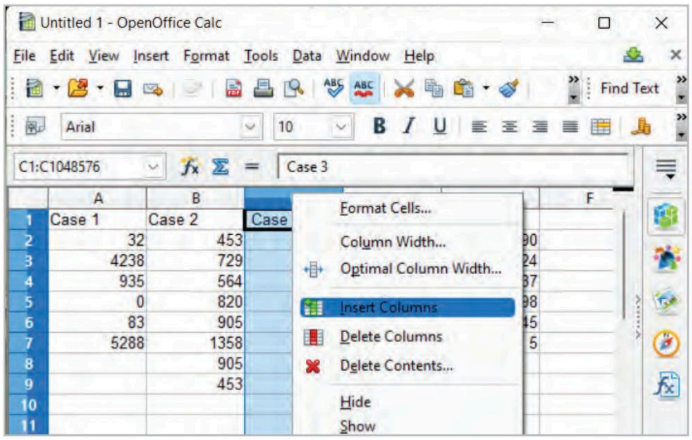


Fig. 11.18: Adding a New Column

Inserting a Row

To insert a row, follow the steps given below:

1. Right-click on the row header where you need to add a new row (Row 6, for example).
2. Select the **Insert Rows** option.

   OR

   click on the **Rows** option from the **Insert** menu.
3. This will shift the selected row to one row down and add a new one to its place. That is, a new row 6 gets added, while the selected row becomes row 7. (See Fig. 11.19.)

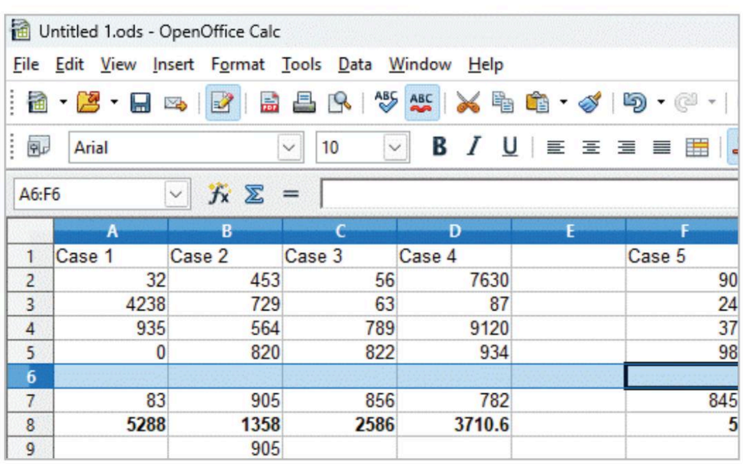


Fig. 11.19: Adding a New Row

Deleting Columns and Rows

To delete the columns or rows, follow the steps given below:

1. Select the column or row to be deleted (Row 6, for example).

   OR

   Right-click on the row header (or the column header).
2. Select the **Delete Rows** (or the **Delete Columns**) option from the drop-down menu.
3. The selected row or column is deleted. (see Fig. 11.20)

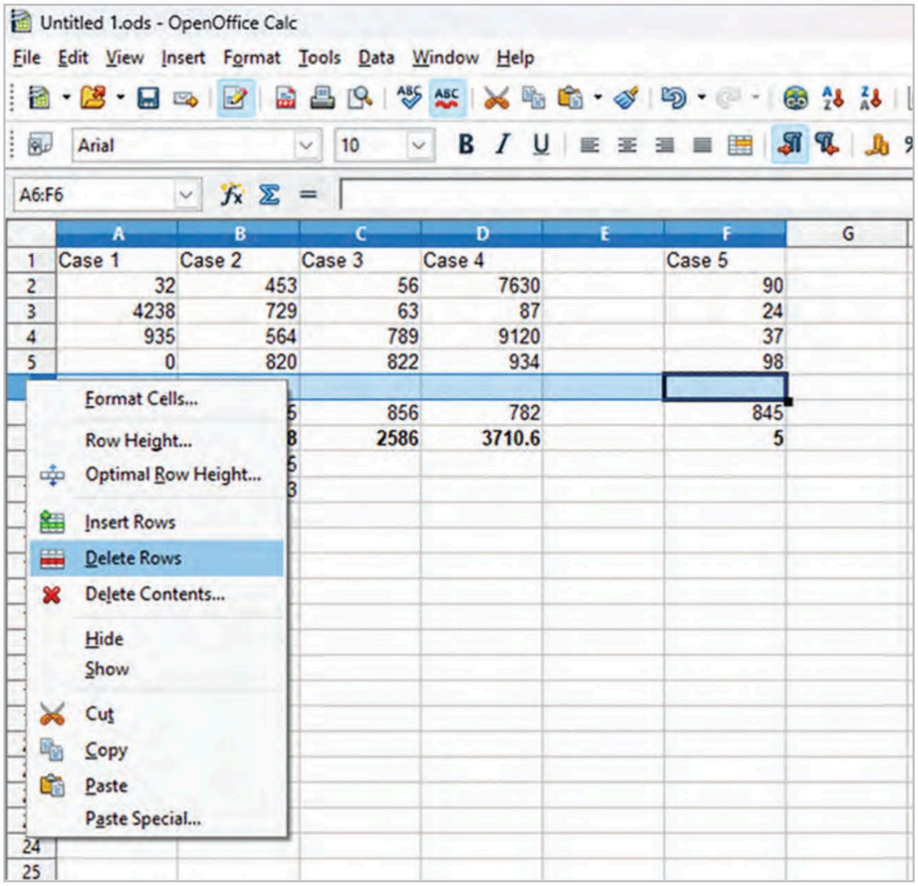


Fig. 11.20: Deleting a Row

Copying and Moving Data

Data entered in a worksheet can be either duplicated or made to change its position within the same worksheet or to a different sheet. The data which is to be moved needs to be cut from the original source whereas the data that needs to be duplicated needs to be copied from the original source. The data either cut or copied is transferred on a temporary memory of the computer, called Clipboard. This data on the clipboard must then be pasted to a location where you want it to move or duplicate. The action can be performed as follows:

Copying Data

The data in the given cells, rows, or columns can be copied as follows:

1. Select the cell(s), row(s), or column(s) whose data needs to be copied.

2. Right-click and select the **Copy** option.

   OR

   Go to the **Edit** menu and select **Copy** from the drop-down menu.

   OR

   Use the keyboard shortcut key **Ctrl+C** (see Fig. 11.21).

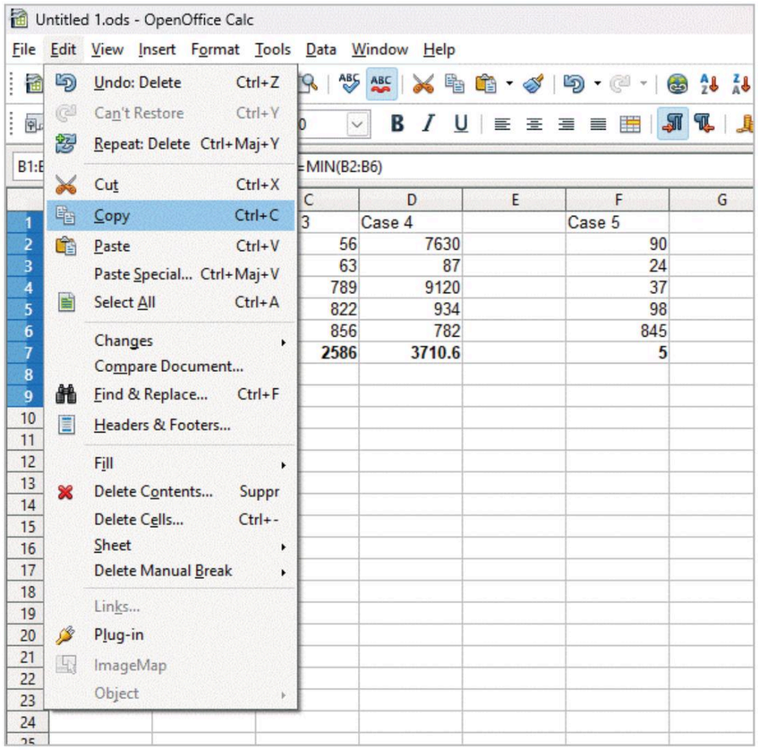


Fig. 11.21: Copying Data From Source

3. Go to the target sheet or cell where the data is to be pasted.
4. Right-click and select the **Paste** option.

   OR

   Go to the **Edit** menu and select **Paste** from the drop-down menu.

   OR

   Use the keyboard shortcut **Ctrl+V**.

Moving Data

The data in the given cells, rows, or columns can be moved as follows:

1. Select the cell(s), row(s), or column(s) whose data needs to be moved.
2. Right-click and select **Cut** option.

   OR

   Go to the **Edit** menu and select **Cut** from the drop-down menu.

   OR

   Use the keyboard shortcut **Ctrl+X** (see Fig. 11.22).

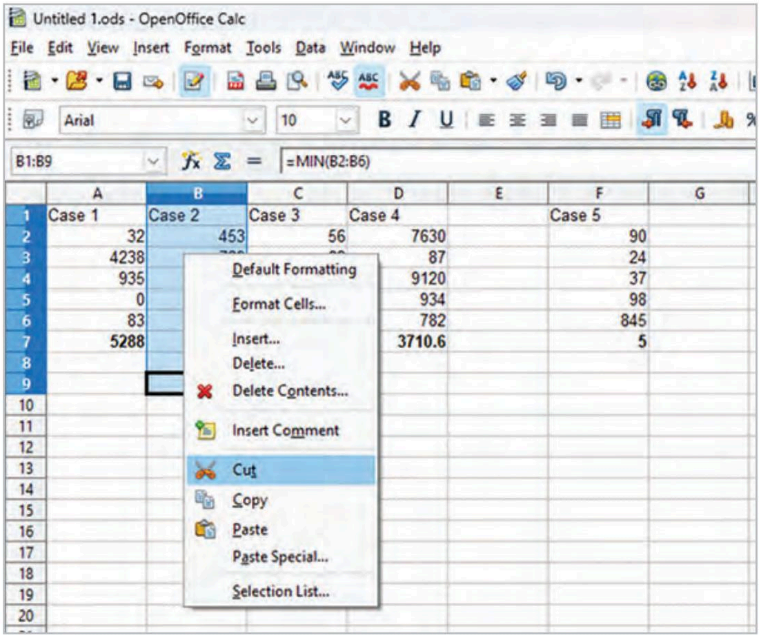


Fig. 11.22: Moving Data From Source

3. Go to the target sheet or cell where the data is to be moved.
4. Right-click and select the **Paste** option.

   OR

   Go to the **Edit** tab and select **Paste** from the drop-down menu.

   OR

   Use the keyboard shortcut **Ctrl+V**.
5. This pastes the data to the selected location.

Using the Drag and Drop Method to Move Data

The data in the given cells, rows, or columns can be moved as follows:

1. Select the source cell, row, or column from where data has to be moved. Here, we have selected the cells under column C.
2. Click the mouse and drag the cells to the desired location, and then release the mouse button. This will shift the data to the new destination (say, Column G), see fig. 11.23.

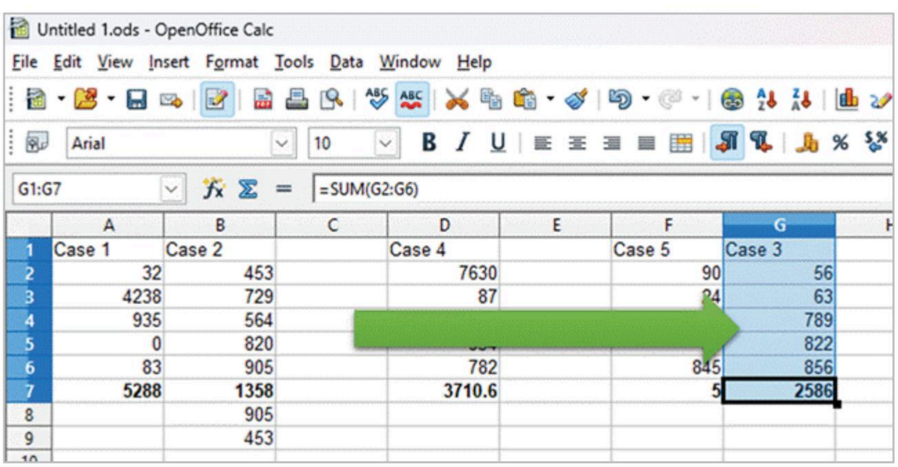


Fig. 11.23: Moving Data by Drag and Drop Menu

Activity Time

Create a spreadsheet with a column labeled "Data" and enter a set of numerical values in cells A2 to A10: ***(Individual Activity)***

5, 11, 8, 19, 6, 10, 7, 13, 9.

Apply the required formulas and functions to:

- find the maximum value in the provided data set.
- find the minimum value in the data set.
- calculate the sum of all the values in the data set.
- find the average of the values in the data set.

Chapter Checkup

A Select the correct option.

1 The expression for a formula starts with .... sign.

a =   b &

c *   d (

2 The values of the data or text which do not change are called ....

a Constants   b Operators

c Numerals   d Functions

3 Which of the following is a shortcut key to move data?

a Ctrl + C and then Ctrl + V   b Ctrl + V and then Ctrl + X

c Ctrl + X and then Ctrl + V   d Ctrl + V and then Ctrl + C

B Fill in the blanks with the most suitable words.

1 Joining two or more strings of text data is called ....

2 .... are the values input to the functions.

3 Calc will recognise the value (204) as ....

4 By default, numbers are .... aligned in a cell.

C State whether the following is *True* or *False*. Correct the statements that are false.

1 The operations which are predefined as formula in Calc are called functions.

2 TODAY( ) function gives us the current date.

3 Mouse handle can be used to drag and fill the adjacent cells with the same formula.

4 To insert a new row, click on the Rows option from the Edit menu.

D Answer the following questions. *(Solved)*

Q1. What is a function?

A1. The operations which are predefined as formula for performing accurate calculations in Calc are called functions. Examples of some commonly used functions are Sum(), Average(), Min(), Max(), etc.

Q2. Evaluate the following equations using operator precedence, and then test the result in the spreadsheet.

a. 8-4/2   **b.** 5*5+8   **c.** 3+5*4   **d.** 2^5+8

e. 3+2^2   **f.** 5+6*2^2   **g.** 8/4*4   **h.** –4/2+2

i. 1+2^2-2   **j.** 4*3/2

A2. **a.** 6 **b.** 33 **c.** 23 **d.** 40 **e.** 7

f. 29 **g.** 8 **h.** 0 **i.** 3 **j.** 6

Q3. Rajat has entered a formula in cell A5 in Calc. He wants to fill the adjacent cells with the same formula as entered in the selected cell. Name the feature of Calc that can help him achieve this.

A3. Fill handle

Answer Key

A **1.** a **2.** a **3.** c

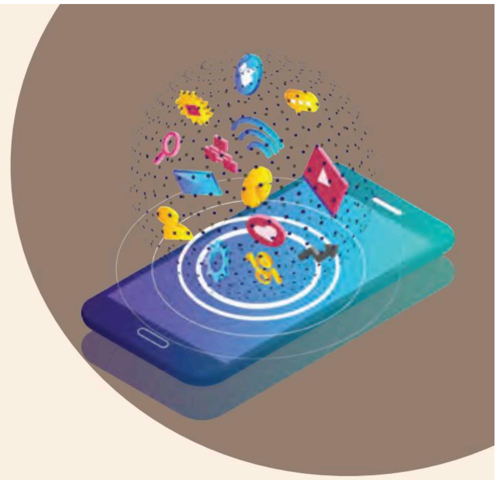
B **1.** Concatenation **2.** Arguments **3.** –204 **4.** Right

C **1.** True.

2. True.

3. False. Fill handle can be used to drag and fill the adjacent cells with the same formula.

4. False. To insert a new row, click on the Rows option from the Insert menu.



12

Formatting Data

Data formatting is the process of changing the appearance and layout of a cell or range of cells to make them easier to read and more visually pleasing. Formatting data in a cell gives additional effect to the text. Additional effects include changing the font style, font size, automatic wrapping, bold, underline, italic, etc. Refer Fig. 12.1 to have a view of various types of formatted cells.

In this chapter, you will learn how to format cells in a worksheet.

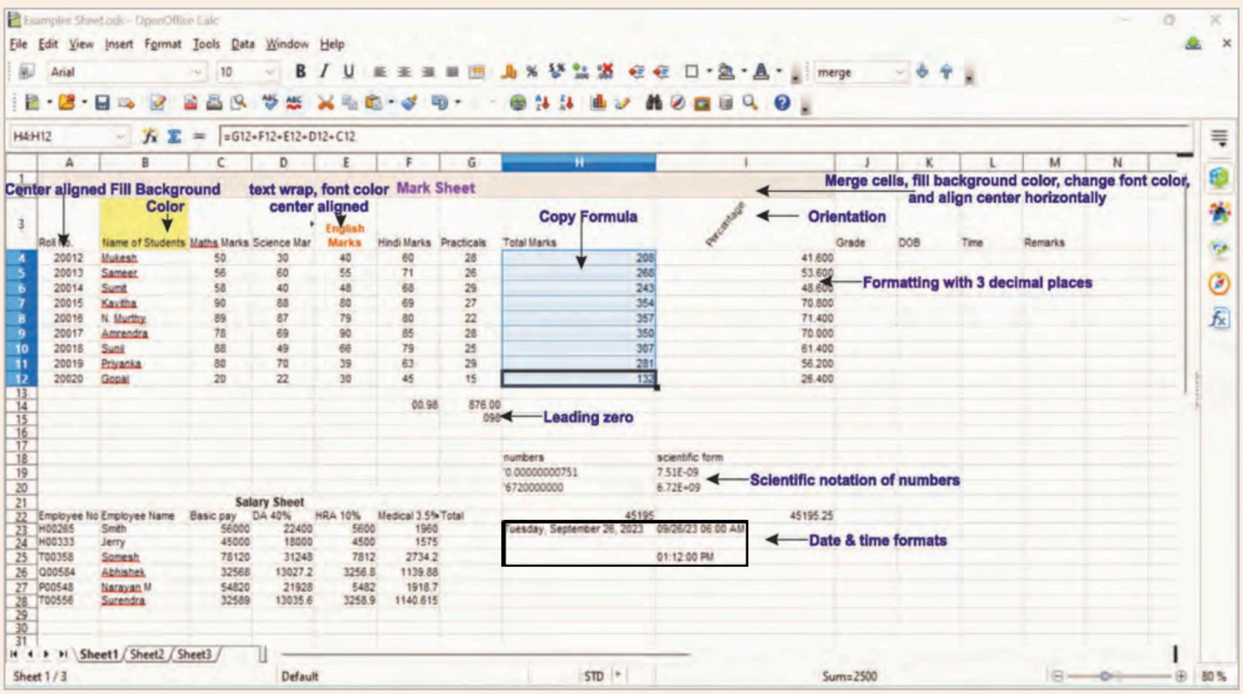


Fig. 12.1: Various Types of Formatted Cells in a Worksheet

Formatting the Worksheet

To format the worksheet, choose the appropriate cell or range of cells where the formatting has to be applied. It can be applied in three different ways.

1. Apply formatting using the tools present on the formatting toolbar. Refer Fig. 12.2.

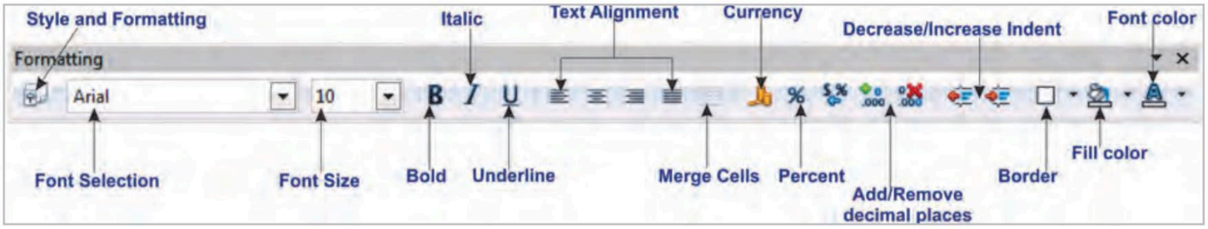


Fig. 12.2: Using Formatting Toolbar

OR

2. Right-click on a cell or range of cells, and choose **Format Cells** option. See Fig. 12.3.

OR

3. Select the desired range of cells. Click on **Format** menu and select the **Cells** option. (See Fig. 12.4.)

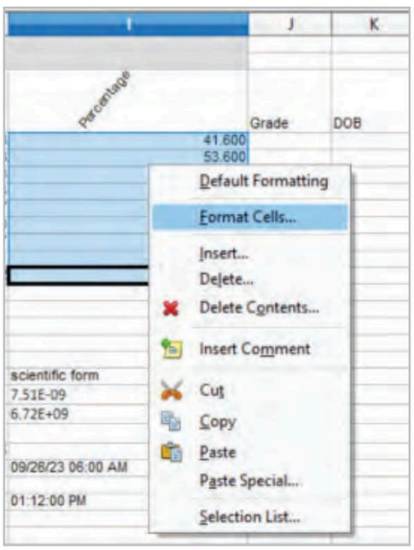


Fig. 12.3: Format Cells Popup Menu

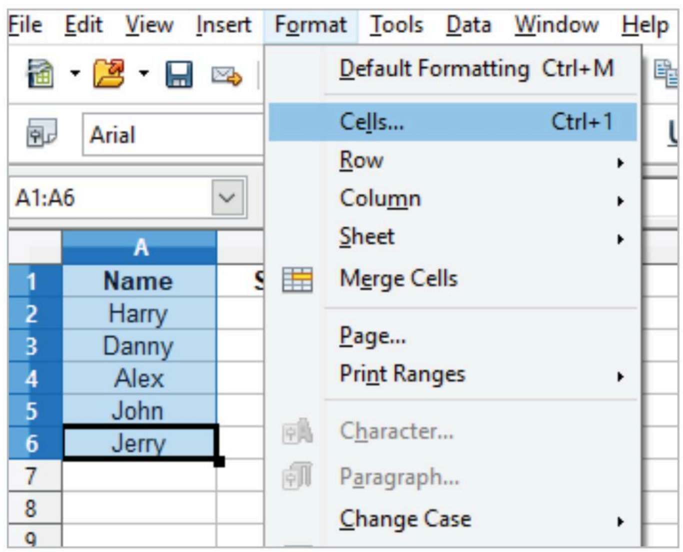


Fig. 12.4: Format Menu

Let's explore the other formatting options that can be used on a worksheet.

Applying Formatting Effects Using Formatting Toolbar

With the aid of the tools on the formatting toolbar, you can change the font family, font style, and font size. You can apply the formatting effects through the following buttons, as shown in Fig. 12.2.

Tools	Descriptions
Font Name	Different font types on a spreadsheet
Font Size	Different font sizes on a spreadsheet
Bold	Make the selected text bold
Italic	Italicises the selected text
Underline	Underlines the selected text
Align Left	Aligns text to the left of a cell
Align Center Horizontally	Aligns text to the centre of a cell

Tools	Descriptions
Align Right	Aligns text to the right of a cell
Justified	Aligns the text evenly on both the left and right margins of a cell
Merge Cells	Combines two or more cells to create a new, larger cell
Borders	Applies a border to the cells
Background Color	Changes the background colour of the cell
Font Color	Makes the text stand out against the white background of the spreadsheet

Formatting Numbers

Although number formats may seem straightforward, using the wrong ones could ultimately result in inaccurate reporting. The context can be better explained with a straight-forward scenario.

Imagine that you work in an automobile business and that your store manager asks you to identify the customer who made the highest expenditure in the month so that you can give him a discount on his subsequent purchase.

In this scenario, imagine that all transactions, except one, were recorded in the standard dollar format. Unfortunately, due to a hurried data entry process, you mistakenly entered the transaction amount for one of the customers in the Indian rupee format. This leads to the wrong analysis of the highest-spending customer. This example conveys how easily errors can arise and how the precision of number formatting plays a critical role in ensuring data accuracy, particularly in important decision-making processes like customer recognition and rewards.

Let's make the number formatting consistent by making all the numeric data in the Indian rupee format.

1. Enter the data in a sheet from cell A1:B6 as shown in Fig. 12.5
2. Select a cell range where you wish to add number formats. In our example, select B2:B6.
3. Click on the **Format** menu and select the **Cells** option. (Refer Fig. 12.4). It will display the **Format Cells** dialog box. (See Fig. 12.6.)
4. Click on the **Numbers** tab. Select a **Category** option. Here, we have selected the **Currency** category.

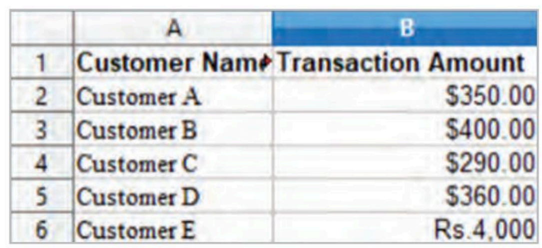


Fig. 12.5: Data Entered

5. Under the **Format** section, select the currency type you would want for your numeric data. Here, we have selected the **INR Rs. English (India)** format.

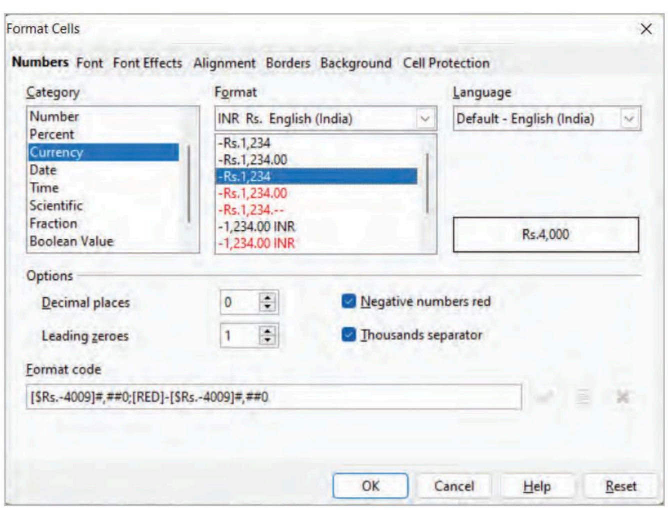


Fig. 12.6: Format Cells Dialog Box

6. Click on the **OK** button.

Number Formats in Calc

The following are the types of number formats available in OpenOffice Calc.

Name	Description
General	General is the default number format selected for any number you type into the spreadsheet. The number appears the same way you type it, without additional decimals or modifications.
Number	The Number format is exclusively used when you are working with the numbers. The number format will add decimal points to your data to keep it more accurate, and you can also customise the number of decimal points you wish to have.
Percent	The Percentage data format represents the values in terms of percentages. For example, 0.20 can be represented as 20%. In this number format too, you can choose the number of decimal places for your percentage value.
Currency	The Currency number format is used when you want to represent numbers in the form of currency. For example, annual investment data. You might use dollars to present such data.
Date	The Date format is used in spreadsheets to consider your input as a calendar date instead of a regular number. For example, 31 December 2023, 31/12/23, 2023-12-31, etc., are different date formats.
Time	The Time format converts the general default number format to the time format. For example, 13:37, 01:37 PM, 13:37:46, etc., are different time formats.
Scientific	The Scientific number format differs slightly from currency or a general number format. The Scientific Number Format has an exponential number that indicates the power of a number. Apart from exponents, the scientific number format also includes trigonometry, calculus, roots of a number, etc.
Fraction	The Fraction format displays the general number in the form of a fraction according to the type of fraction you chose.
Boolean Value	Boolean variables are stored as 16-bit (2-byte) numbers, but they can only be true or false.
Text	The text includes the textual type of data. If number data is typed in text format, then Calc keeps it the way it is without adding any decimals or mathematical symbols to the input number data.
User-defined	Apart from all the available options, OpenOffice Calc also provides its users with a customisable option that will help them use existing formats and customise them to create a completely new one.

Formatting Cells as Numbers

Consider the format of the numeric data in the given table representing the Science and Maths marks of various students.

D	E	F
Name	**Science Marks**	**Maths Marks**
Harry	13	12
Danny	98	45
Alex	72	76
John	68	87
Jerry	90	56

Fig. 12.7: Numeric data

To apply the number format, follow the given steps:

1. Select a cell range where you wish to add number formats.
2. Click on the **Format** menu and select the **Cells** option. It will display the **Format Cells** dialog box.

3. Select the **Number** format that is to be applied to the data.
4. In the **Decimal places** option, select the number of decimal places you want to display in the data.
5. Indicate the number of leading zeroes by choosing a suitable value in the **Leading zeroes** spin box.
6. Click on the checkbox for the **Thousands separator** to use a comma as a separator between thousands of values. This command is optional.
7. Click **OK**. The format has now undergone the following modification, as shown in Fig. 12.8.

Name	Science Marks	Maths Marks
Harry	13.00	12.00
Danny	98.00	45.00
Alex	72.00	76.00
John	68.00	87.00
Jerry	90.00	56.00

Fig. 12.8: Changing Number Format

Formatting a Range of Cells as Labels

Numerous numeric values beginning with zero ('0') are required to be entered while storing data in a spreadsheet, including postal codes, STD/ISD codes, phone numbers, employee codes, etc.

However, when entering such a number, the initial zero ('0') vanishes. This is due to the fact that these are kept as a numeric value, and a zero does not come before a numeric number. If you want to display these numerical values with a leading zero, then this type of data can be formatted as text or labels.

1. Select the range of cells from B2 to B6.
2. Click **Format**>**Cells** from the menu options. This opens up a **Format Cells** dialog box.
3. Click on the **Numbers** tab.
4. Select **Text** in the drop-down list of the **Category** section.
5. Click the **OK** button.

	A	B
1	Name	Employee Code
2	Harry	001
3	Danny	002
4	Alex	003
5	John	004
6	Jerry	005

Fig. 12.9: Formatting a Range of Cells as Labels

After formatting the data as text, you can manually add as many leading zeros as you want. From the Fig. 12.9, you can see the number of leading zeroes in the numeric data under column B.

Note that the data formatted as labels is left aligned.

Formatting a Range of Cells as Scientific Form

When a number is too big or too small to be stated in a decimal form, or if doing so would entail writing down an unwieldy, lengthy string of digits, it can be expressed using scientific notation. It is also known as standard index form or scientific form.

Scientists, mathematicians, and engineers frequently utilise this ten base notation because it can make some mathematical operations simpler. It is typically referred to as the "SCI" display mode on scientific calculators.

To display/represent numbers as scientific, follow the given steps:

1. Select the range of cells B9:B12.
2. Click **Format**>**Cells** from the menu**.** This opens up a **Format Cells** dialog box.
3. Click on the **Numbers** tab.
4. Select **Scientific** in the drop-down list of the **Category** section.
5. Click the **OK** button.

8	Decimal Notation	Scientific form
9	8	8.00E+000
10	1500	1.50E+003
11	543.78	5.44E+002
12	0.8	8.00E-001

Fig. 12.10: Scientific Numbers

Few examples of scientific numbers is displayed in the following table.

Decimal Notation	Scientific Notation	Displayed in Calc Spreadsheet
2	2×10^{0}	2.00E+000
300	3×10^{2}	3.00E+002
4321.768	4.321768×10^{3}	4.32E+003
−53000	-5.3×10^{4}	-5.30E+004
6720000000	6.72×10^{9}	6.72E+009
0.2	2×10^{-1}	2.00E-001
987	9.87×10^{2}	9.87E+002
0.00000000751	7.51×10^{-9}	7.51E-009

Formatting a Range of Cells to Display Date and Time

Even though dates and time are actually stored as a regular number known as the date serial number, we can make use of extensive OpenOffice date and time formatting options to display them just the way we want.

For various time zones, Calc offers numerous predefined dates and time formats. Calc stores time and dates as decimal values that are divided into an integer for the date and a decimal for the time.

Every date is saved as an integer, indicating how many days have passed since January 1, 1900.

Time entries are saved as decimals between .0 and .99999, where .0 is 00:00:00 and .99999 is 23:59:59, to represent that portion of the day.

Formatting Text

Changing the font of the text in your spreadsheet is an easy way to change its appearance. A font is a collection of letters, numerals, and punctuation symbols that share a common appearance. A font will contain size and style changes. Let us understand them one by one.

Text Alignment

In OpenOffice Calc, alignment is a cell formatting attribute that influences how the data in the cell(s) appears.

There are four types of horizontal alignments: left, center, right, justified, and filled.

1. **Align Left:** Left alignment indicates that the text/values will begin at the left edge of the cell(s).
2. **Align Center Horizontally:** The text and values will be centred in the cell(s) when using the center alignment.
3. **Align Right:** Right alignment indicates that the text/values will begin at the right edge of the cell(s).
4. **Justified:** Justified alignment indicates that the text and values will be aligned on both the left and right edges.
5. **Filled:** The Filled option fills the entire cell area by repeating the contents in it.

There are three types of vertical alignment for text:

1. **Top:** Aligns the text/values at the top of the cell(s).
2. **Middle:** Aligns the text/values at the centre of the cell(s).
3. **Bottom:** Aligns the text/values at the bottom of the cell(s).

Did You Know?

In a cell, the text data is by default aligned to the left. Additionally, by default, numerals are aligned to the right.

Let us understand text alignment with the help of an example.

1. Enter the data as shown (See Fig. 12.11)
2. Place the cursor in cell **B4,** whose alignment you wish to change.
3. Click and drag to select multiple cells till **B12**.
4. Right-click on the selected area and select the **Format Cells** option to open the **Format Cells** dialog box.
5. Click on the **Alignment** tab.
6. In the **Text alignment** section, from the **Horizontal** drop-down list, select **Left**, **Center**, **Right**, **Justified**, or **Filled** as desired. Here, we have selected **Center**.
7. From the **Vertical** drop-down list, select **Top**, **Middle**, or **Bottom**. In this example, we have selected **Middle**. (See Fig. 12.12.)
8. Click on **OK**.

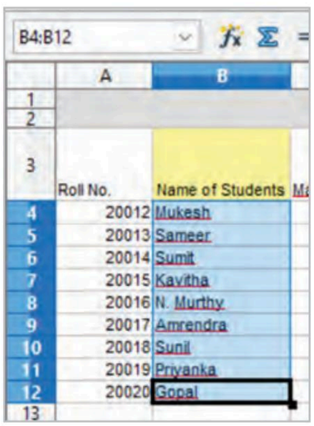


Fig.12.11: Cells to be Aligned as Center

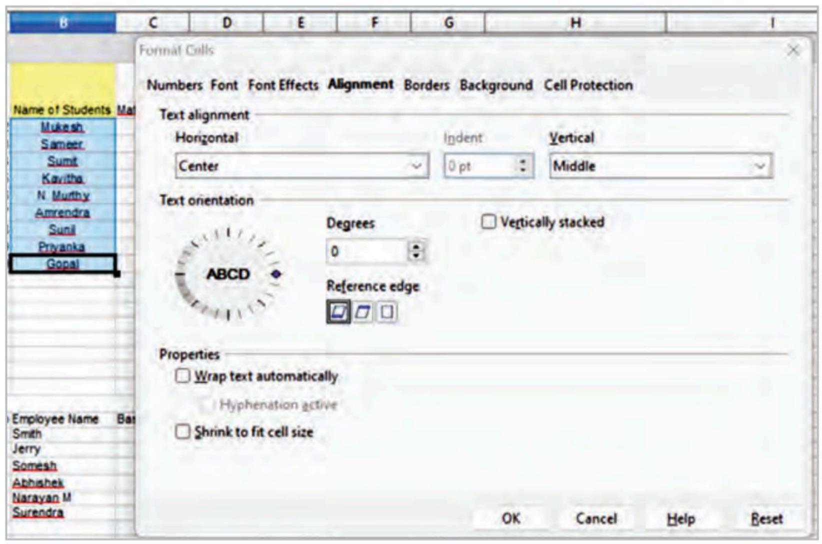


Fig. 12.12: Selecting Text Alignment

Text Orientation

Orientation helps rotate the text values diagonally or vertically.

The orientation of the cell is a significant formatting characteristic that describes the direction in which the contents of the cell are to be displayed or printed.

You may rotate text in a cell or group of cells in many different ways using **Degrees**.

For the selected cell(s), enter the rotation angle or the degree to which you want to rotate the text. If you click the upward pointing arrowhead of the **Degrees** spinbox, it will cause the text to rotate left, while the downward pointing arrowhead will cause the text to spin to the right.

Reference Edge

Select the **Reference Edge** option in the **Format Cells** dialog box's **Text orientation** section, to display the cell edge to write the rotated text. (See Fig. 12.13.) Reference Edge has three options:

Text Extension From Lower Cell Border—displays the rotated text from the bottom cell edge onwards. (Cell A1)

Text Extension From Upper Cell Border—displays the rotated text from the top cell edge outwards. (Cell B1)

Text Extension Inside Cell—displays the rotated text within the cell. (Cell A2)

The **Vertically stacked**—option stacks the letters on the top of each other. (Cell B2)

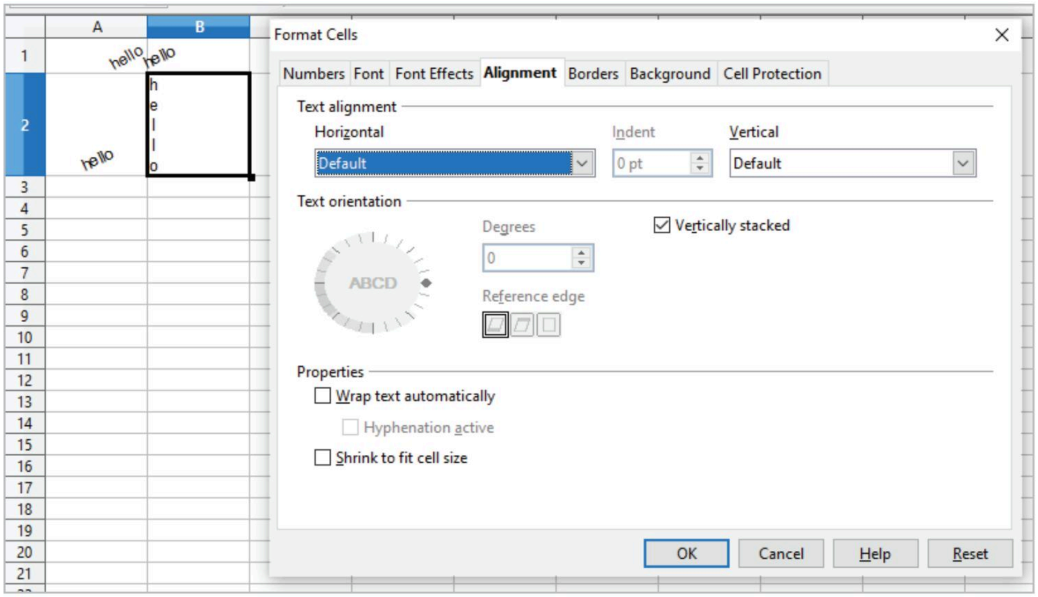


Fig.12.13: Illustration of Cell Orientation

Did You Know?

Vertically stacked option is only available if you enable support for Asian languages in **Tools** > **Options** >**Language Settings** > **Languages.**

Remember

The **Properties** section under the **Alignment** tab provides options to change the text flow in a cell.

Wrap Text Automatically

The best approach to fitting a lot of text into OpenOffice Calc cells is to use the text wrapping option. It allows you to rapidly change the cell's dimensions so that all the text neatly fits inside it rather than extending outside the cell's boundaries. The width of the cell determines how many lines will fit in.

Did You Know?

To enter a manual line break, press the **Ctrl+Enter** key combination in the cell.

1. Select the cell whose text needs to be wrapped.

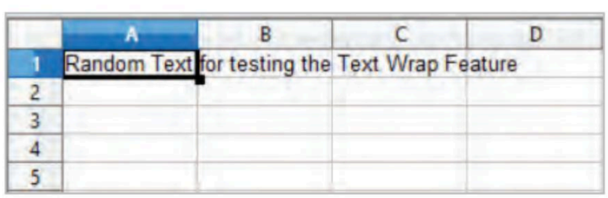


Fig. 12.14: Text Before Applying the Wrap Text Feature

2. Right-click the selected cell.
3. Go to the **Format Cells** popup option, or select the **Cells** option from the **Format** menu. The **Format Cells** dialog box will open. (See Fig. 12.13.)
4. Click on the **Alignment** tab.
5. Check the **Wrap text automatically** box in the **Properties** section.
6. Click **OK** to finish.

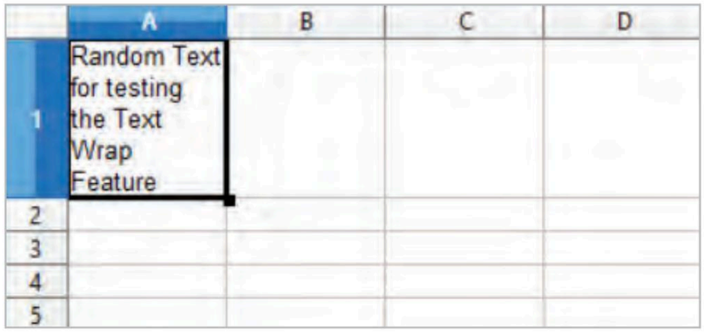


Fig. 12.15: Text After Applying the Wrap Text Feature

Hyphenation Active

Word hyphenation is enabled for text wrapping. Hyphenation in text wrapping is a feature that automatically adds hyphens to break words at the end of lines when text is being wrapped.

Shrink to Fit Cell Size

Reduces the font's perceived size so that the cell's contents fit within the available cell width.

> **Did You Know?**
>
> Shrink to fit cell size cannot be used on a cell that has line breaks.

The Fill Handle

Although entering data into a spreadsheet can be highly laborious, OpenOffice Calc offers a number of features to make it easier. The fill handle is one such feature. It is a small black box at the bottom right corner of the cell. It automates input, particularly repetitive data or data that has patterns.

Using the Fill Handle

The OpenOffice Calc Fill Handle tool is used to fill the subsequent cells with the predefined value by dragging it to the desired cells. For instance, to fill in the numbers 1, 2, 3, or the days of the week Monday, Tuesday,..., or the name of the month January, February,..., enter the first two values in the cells, select them, then drag them to the following cells till you want to continue the series in succession.

Let us understand it with the help of an example. (See Fig. 12.16.)

For example, if you want to generate the whole number series 1, 2, 3,... up to 22 using the Fill handle tool, follow the given steps:

1. In cell A1, type 1 and press the **Enter** key.
2. Click A1 to place the cell pointer.
3. Click the cell's Fill Handle on the bottom-right corner of the cell. You will notice that the mouse cursor changes to a little plus sign (+).
4. While dragging over the cells, the generated values will be presented. Stop dragging when you reach the desired number, 22.

You will notice a series of numbers is generated automatically.

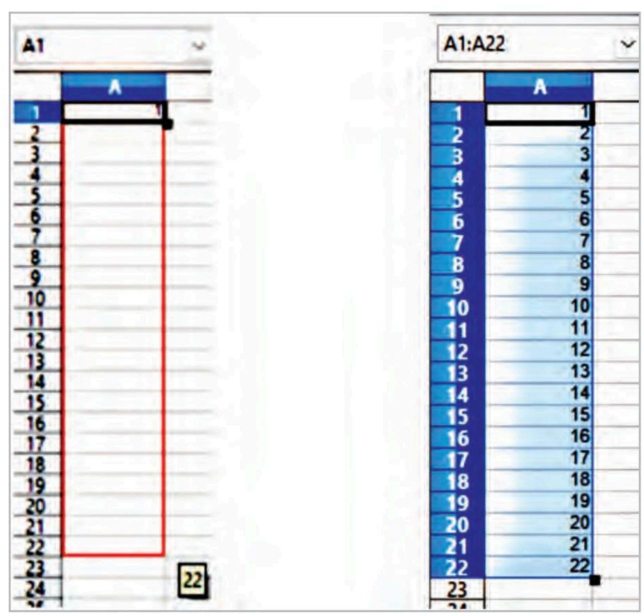


Fig. 12.16: Illustration of Fill Handle

Remember

You can also generate a series at different intervals. To get an odd number series, for example, type 1 in the first cell and 3 in the next, select the two cells, and drag the fill handle up to the desired number of cells.

Did You Know?

1. To fill cells quickly, grab the "handle" in the bottom right-hand corner and move it in the direction you wish to fill. If the cell includes a number, it will be filled in series. If the cell includes text, the content will be filled in the direction you selected.
2. If you do not wish to fill the cells with different values, hold down the Ctrl key. This key will ensure that a repetitive series is generated instead of a series based on a pattern.

Use of a Fill Handle to Copy a Formula

If you want to apply the same formula to a number of cells, you don't need to manually input the formula repeatedly in each cell. You can instead use the fill handle to copy the formula. This saves time and effort and lowers the possibility of mistakes, especially when dealing with lengthy and complex formulas.

This is an alternative method to copying and pasting. The AutoFill feature copies the contents of one cell to all dragged cells. The content could be data or a formula. When you enter a formula, the addresses of all filled formulas are modified.

Follow the steps below to copy the formula using the fill handle.

1. Click the H4 cell, which contains the formula for total marks. A small black + sign (fill handle) appears in the bottom-right corner of the selected cell.
2. Drag the fill handle to cell H12.
3. Release the mouse button. The total marks of all respective students will be displayed automatically. (See Fig. 12.17.)

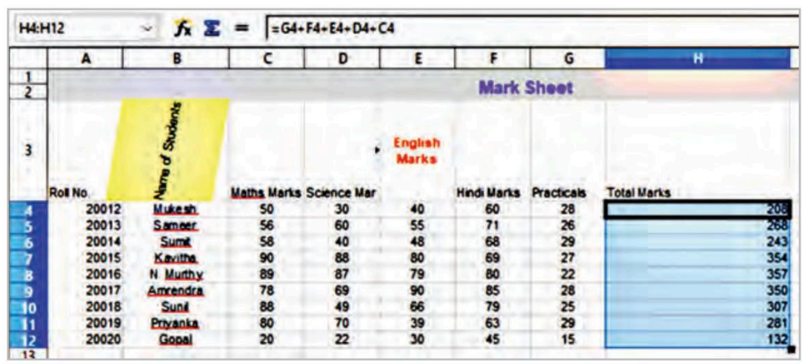


Fig. 12.17: Copying Formula With Fill Handle

Using a Selection List

Only text can be used in selection lists. The selection list suggests text that has already been typed in the same column.

Click a blank cell and press **Ctrl+D** to use a selection list. A drop-down list appears for any cell in the same column that has at least one text character. Select the required entry.

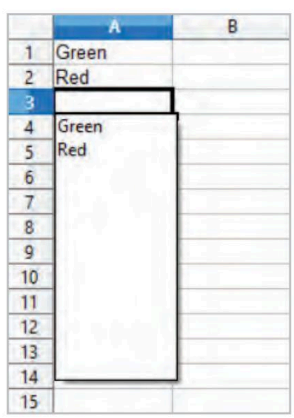


Fig. 12.18: Using a Selection List

Activity Time

1. Assume that you have the given data in cell A1, as shown in Fig. a. You want to display text as given in Fig. b. How will you achieve this?

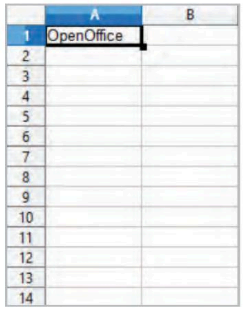


Fig. a

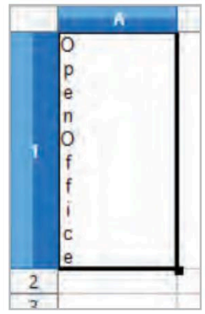


Fig. b

2. Identify the feature of Calc used so that the text in cell A1 of Fig. c appears as that in Fig. d.

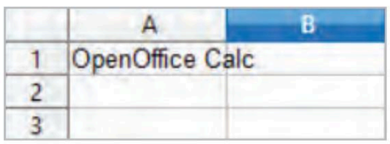


Fig. c

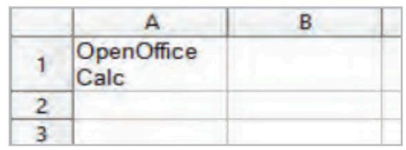


Fig. d

3. Name the feature of Calc used to obtain the data as given in Fig. f.

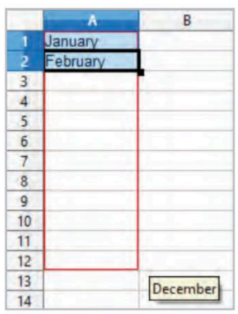


Fig. e

Fig. f

Chapter Checkup

A **Select the correct option.**

1. Numbers entered into a cell are automatically:
   - a) Left aligned
   - b) Right aligned
   - c) Center aligned
   - d) Justified

2. The Autofill feature in a spreadsheet:
   - a) Generates a sequential series of data
   - b) Changes the orientation of the selected text
   - c) Applies a border around the selected cells
   - d) Fills the selected cells with a background colour

3. Which of the following is a type of number format available in OpenOffice Calc?
   - a) Percent
   - b) Currency
   - c) Date
   - d) All of these

B **Fill in the blanks with the most suitable words.**

1. .... is the process of changing the appearance and layout of a cell to make it easier to read and visually pleasing.
2. .... helps rotate the text values diagonally or vertically.
3. .... allows you to change the cell's dimensions, so that all the text neatly fits inside it rather than extending outside the cell's boundaries.
4. .... alignment aligns the data evenly on both the left and right side of a cell.

C **State whether the following is *True* or *False*. Correct the statements that are false.**

1. Only numbers can be used in selection lists.
2. Merging cells combine two or more cells to create a new, larger cell.
3. Every date is saved as a text indicating how many days have passed since January 1, 1900.
4. The Formatting toolbar has an option to change the font size of the text.

D **Answer the following questions. *(Solved)***

Q1. What is text wrapping?

A1. The text wrapping option allows you to rapidly change the cell's dimensions so that all the text neatly fits inside it rather than extending outside the cell's boundaries.

Q2. What is orientation?

A2. Orientation helps rotate the text values diagonally or vertically. The orientation of the cell is a significant formatting characteristic that describes the direction in which the contents of the cell are to be displayed or printed.

Q3. If the values in the cells are as given: A1 = 5, A2 = 10, and A3 = 15, then to continue the series up to A20, which tool can we use?

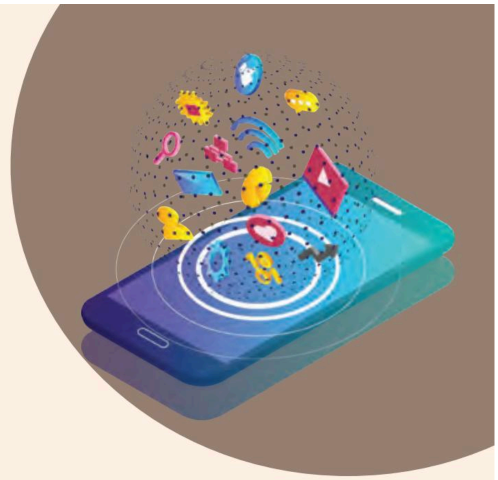
A3. We can use the Fill handle tool of the Calc to generate the series 5, 10, 15, ... 100.

Answer Key

A **1.** b **2.** a **3.** d

B **1.** Formatting **2.** Orientation **3.** Text wrapping **4.** Justify

C **1.** False. Only text can be used in selection lists.
2. True.
3. False. Every date is saved as an integer indicating how many days have passed since January 1, 1900.
4. True.



13

Cell Referencing

Each worksheet in an OpenOffice Calc consists of several cells formed by the intersection of rows and columns. Each cell has a specific reference (or cell address). For example, the cell address "A10" refers to the cell in the tenth row of column A. The cell reference helps the users to easily refer to the desired cell(s) within the functions or formulas.

This chapter provides a basic overview of OpenOffice Calc cell references. The chapter also goes through the various sorts of cell references accessible in OpenOffice Calc, as well as the step-by-step techniques for using them.

What Is Cell Referencing?

Cell referencing refers to a cell or range of cells on a spreadsheet and can be used in the formula so that the values or data associated with that cell can be utilised for calculation in formulas. We must supply the column name followed by the row number of the appropriate cell when representing the cell reference. The figure below (see Fig. 13.1(a)) shows the cell reference of the selected cell in an OpenOffice sheet:

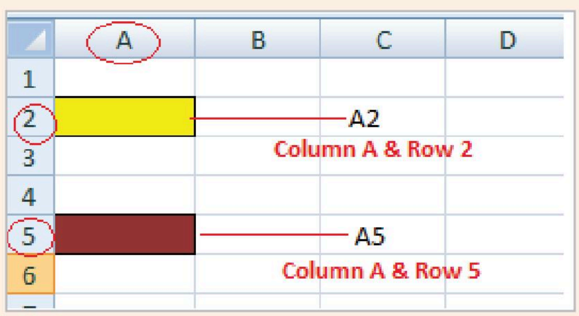


Fig. 13.1(a): Cell Reference of the Selected Cell

The cell reference assists in locating the cell of the sheet and reading or using its data in the required formula or function to generate the result. The cell being referred can be on the same sheet, on a different sheet, or on a separate workbook.

External referencing is used when referencing cells from other workbooks. Remote reference occurs when cells are referenced from other spreadsheet applications.

Using a Simple Cell Reference

Let us now look at some easy examples of how to use a simple cell reference:

Simply specifying the referred cell with the equal sign demonstrates the basic use of a cell reference. For example, if we type "=A1" without quotation marks in another cell on the sheet, the value in cell A1 will be displayed in the relevant cell. This signifies that the value of the selected cell, where the cell reference is entered, is the same as cell A1 (see Fig. 13.1(b)).

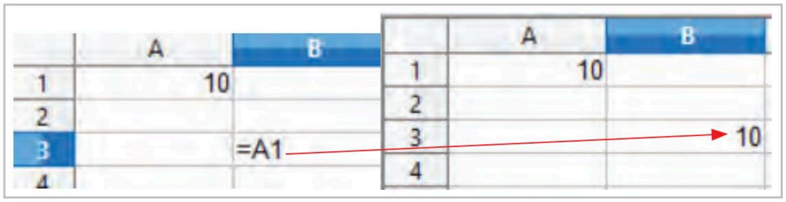


Fig. 13.1(b): Cell Reference

Reference to a Cell Range

If we use the notation "=A2:C6" without the quotes, we refer to the entire cell range from cell A2 to cell C6. However, a range alone is not valuable data in Calc. When we use this cell reference in Calc, it gives the #VALUE! error, which indicates that the formula is missing (see Fig. 13.2).

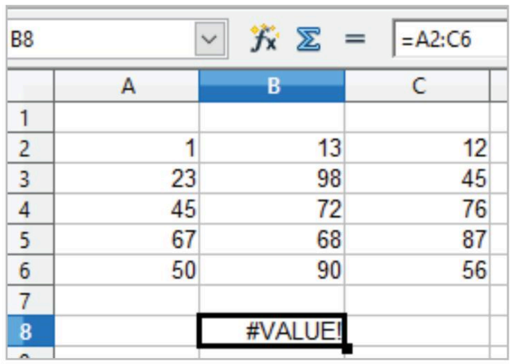


Fig. 13.2: Reference to a Cell Range

Reference to a Cell Range in a Function

You can utilise reference to a range in a function. For instance, =SUM(A2:C6) will return the sum of the values in the cells, whereas =AVERAGE(A2:C6) will provide the average of the values in these cells (see Fig. 13.3).

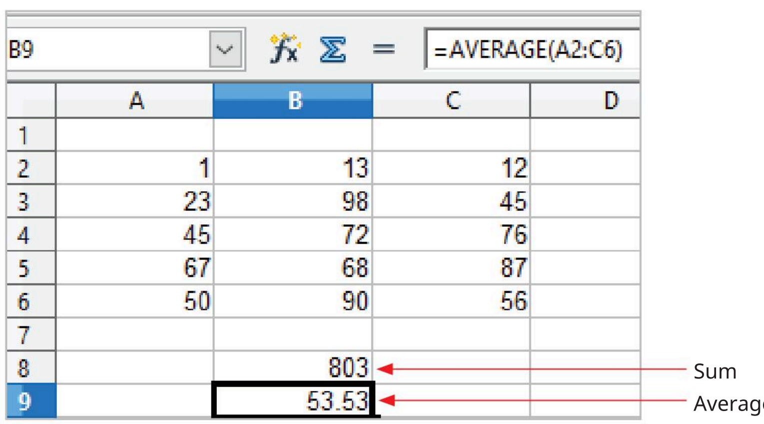


Fig. 13.3: Reference to a Cell Range in a Function

Types of Cell Reference in OpenOffice Calc

Cell referencing is really helpful when copying and pasting OpenOffice Calc formulas in multiple cells. OpenOffice Calc provides three major types of cell references based on diverse use scenarios, including:

1. Relative cell reference
2. Absolute cell reference
3. Mixed cell reference

Relative Cell Reference

OpenOffice Calc by default, uses a relative cell reference. When we include a cell reference or a range in a formula, OpenOffice Calc always uses a relative reference. The associated cell references are used in conjunction with the relative references, which often indicate the combination of column name and row number. The relative reference for the cell does not contain a dollar ($) sign.

Use of Relative Cell References in OpenOffice Calc

Let's see how to use the relative cell references. The below-mentioned "Marks Analysis" table contains "Name" in column A (A2:A6), "Science Marks" in column B (B2:B6), "Maths Marks" in column C (C2:C6) and "Total Marks" in column D, whose values we need to find out (see Fig. 13.4).

	A	B	C	D
1	**Name**	**Science Marks**	**Maths Marks**	**Total Marks**
2	Harry	13	12	
3	Danny	98	45	
4	Alex	72	76	
5	John	68	87	
6	Jerry	90	56	

Fig. 13.4: Marks Analysis

To calculate the Total Marks for each student, we need to add the Science Marks and Maths Marks of each student. So, the formula in cell D2 would be =B2+C2.

Now, instead of entering the formula for all the cells, we can apply the above formula to the entire range. Follow the given steps to copy the formula down in column D:

1. Select cell D2.
2. Scroll down the mouse from the corner, as shown below, until cell D6. This is the AutoFill feature.

3. The total marks for cell D3 becomes '=B3+C3', for cell D4 becomes '=B4+C4', and so on; this is known as **relative referencing**.

Here, when you copy or move a formula with a relative cell reference to another row, automatically, row references will change (similarly for columns also).

To check a relative reference, we must select any of the "Total Marks" cells in column D, and we can view the formula in the Input line (formula bar). E.g., in cell D6, we can observe that the formula has been changed from =B2+C2 to =B6+C6 (see Fig. 13.5).

D6 | =B6+C6

	A	B	C	D
1	**Name**	**Science Marks**	**Maths Marks**	**Total Marks**
2	Harry	13	12	25
3	Danny	98	45	143
4	Alex	72	76	148
5	John	68	87	155
6	Jerry	90	56	146

Fig. 13.5: Relative Reference

When to Use Relative Cell References

Relative cell references are useful when you need to create a formula for a group of cells, and the formula needs to make a reference to its relative cells.

If so, you can create the formula in one cell, copy it, and then paste it into each subsequent cell. You can also use the AutoFill feature of Calc for the same.

Absolute Cell Reference

There are situations when the cell reference must remain the same while copying or utilising AutoFill. Dollar signs are used to maintain a continuous column and/or row reference. Therefore, we can employ the dollar sign ($) characters to convert a relative reference into an absolute reference. When you place the $ symbol before the column letter and the row number, you obtain an absolute column or row.

We utilise absolute reference to refer to a specific fixed point on a worksheet whenever copying is performed. Since the reference is locked, when it is copied, the rows and columns won't change.

How to Use Absolute Cell Reference?

Below is an example depicting how to use absolute cell references.

The below-mentioned Items_on_sales table contains "Items" in column A (A2:A6), "Old_Price" in column B (B2:B6), and "New_Price" in column C, whose values we need to find out with the help of an absolute cell reference.

We can see that the rate for each product has increased by 5% effective from January 2023 and this rate is listed in cell "A9" (see Fig. 13.6).

	A	B	C
1	**Items**	**Old_Price**	**New_Price**
2	Paint	500	
3	Duster Cloth	75	
4	Fevicol	200	
5	Paint Brush	150	
6	Primer	300	
7			
8	**% of rate increase effective from Jan 2023**		
9	5		

Fig. 13.6: Items_on_Sales

To calculate the "New_Price" for each item, we need to multiply the old price of each item by the percentage price increase (5%) and add the "Old_Price" to it.

The formula to calculate the new price for item Paint in cell C2 would be =B2*$A$9+B2.

Here, the percentage rate increase for each product is 5%, a common factor. Therefore, we must add a dollar symbol in front of the row and column number for the cell "A9" to make it an absolute reference, i.e., $A$9. Here the dollar sign fixes the reference in such a way that it remains unchanged even when you copy the formula to other cells.

When to Use Absolute Cell Reference?

Absolute cell references are useful when you don't want the cell reference to change as you reproduce formulas. This is possible if you have to use a fixed value in the formula.

Mixed Cell Reference

A mixed cell reference consists of an absolute column and a relative row, or a *relative column and an absolute row*. When you place the $ symbol before the column letter or the row number, you obtain a mixed cell reference. $B8 is relative to row 8 but absolute to column B, whereas B$8 is relative to column B but absolute to row 8.

The dollar ($) before the row number fixes/locks the row, and the dollar ($) before the column name fixes/locks the column.

Let's see how to use the mixed cell references through examples. In the below-mentioned table, we have values in each row and column. Here, we have to multiply cell values of each column by each row with the help of a mixed cell reference (see Fig. 13.7).

	A	B	C	D
1		2	3	4
2	7			
3	8			
4	9			

Fig. 13.7: Mixed Reference

We can use two mixed-cell references here to get the desired output.

Let us apply two types of mixed references in cell "B2".

The formula in "B2" would be =$A2*B$1.

1. **$A2** refers to the **absolute column and relative row**.

The dollar sign before column A indicates that only the row number can change. At the same time, the column letter A is fixed. It does not change.

When we copy the formula to the right side, the reference will not change because it is locked, but when you copy it down, the row number will change because it is not locked.

2. **B$1** refers to the **absolute row and relative column**.

The dollar sign before the row number indicates that only the column letter B can change. Whereas the row number is fixed, it does not change.

When you copy the formula downwards, the reference will not change because it is locked. But when we copy the formula to the right side, the column alphabet will change because it is not locked.

In the above example, the formulas in cells will be as shown below.

Cell B2	=$A2*B$1	Cell C2	=$A2*C$1	Cell D2	=$A2*D$1
Cell B3	=$A3*B$1	Cell C3	=$A3*C$1	Cell D3	=$A3*D$1
Cell B4	=$A4*B$1	Cell C4	=$A4*C$1	Cell D4	=$A4*D$1

Instead of entering the formula for all the cells, we can apply a formula to the entire range; see the output in Fig. 13.8 below.

	A	B	C	D
1		2	3	4
2	7	14	21	28
3	8	16	24	32
4	9	18	27	36

Fig.13.8: Result of Mixed References

Remember

The cell reference is a key element of the formula or OpenOffice Calc functions.

Activity Time

Activity 1: Using Absolute Cell Referencing ***(Individual Activity)***

In your spreadsheet, enter ten numbers in sequence, using the AutoFill feature, starting cell A1 upto cell A10. Also enter a random value in cell G1. Create a formula using absolute reference to multiply the constant value in cell G1 with each of the numbers in column A and see results.

Activity 2: Group Discussion ***(Group Activity)***

Divide your classmates into a group of 4-5 students and discuss specific examples to illustrate each type of reference and their practical applications.

Activity 3: Compare Marks Using Cell Referencing ***(Individual Activity)***

Create a spreadsheet with data containing the marks for two semesters of all the students in your class. Create a formula using relative references to calculate the change in marks from the previous semester. Copy the formula to analyse how relative referencing adapts to different columns.

Chapter Checkup

A Select the correct option.

1. Which formula has the absolute cell reference?
   - a =A1+$B$2
   - b =A1*B2
   - c =A1+B2
   - d =A1/B2
2. In the formula, which references will be updated when =$D$3+$E5 is copied?
   - a Column D
   - b Column E
   - c Row 3
   - d Row 5
3. In the formula, =$D3+E5, which reference is absolute?
   - a Column E
   - b Column D
   - c Row 5
   - d Row 3
4. Which formula below contains a relative cell reference?
   - a $A$1+$B$2
   - b $A$3+$B$3
   - c $A$1+$B$3
   - d A1+B2

B Fill in the blanks with the most suitable words.

1. Unlike .... references, ... references do not change when copied or filled.
2. $B$17 is a .... type of reference.
3. .... symbol is used to designate an absolute cell reference.
4. In the formula =$D$3+E$5, ... column is relative, meaning it will be updated if copied or if AutoFill is used.

C State whether the following is *True* or *False*. Correct the statements that are false.

1. Cell value references should always be used to avoid having to make multiple manual updates when one cell value changes.
2. The =F18+F19/2 formula uses relative cell reference.
3. We can employ the percent sign (%) before characters to convert a relative reference into an absolute reference.
4. $A2 refers to an absolute column and a relative row.

D **Answer the following questions.** ***(Solved)***

Q1. What is Cell Reference?

A1. Cell Reference in OpenOffice Calc is an alphanumeric value, which means it consists of a combination of a column letter and a row number that identifies a specific cell in a worksheet. It is used to refer to or manipulate data within that cell.

Q2. Differentiate between absolute and mixed references.

A2. Mixed cell references have dollar signs attached to either the column letter or the row number in a reference but not both (i.e., $A1 and A$1). Absolute cell references have dollar signs attached to both column letter and row number in a reference (for example, $A$1).

Q3. Suyash is working in a Calc spreadsheet where data is spread across various columns and rows. He wants to refer to only a single cell, A1 in all the formulas he is using. But as he copies the formula to different cells, he finds that the cell address keeps on changing. What should he do in such a scenario?

A3. He should use absolute cell reference $A$1 in all the formulas.

Answer Key

A **1.** a **2.** d **3.** b **4.** d

B **1.** Relative, Absolute **2.** Absolute **3.** $ **4.** E column

C **1.** True.

2. True.

3. False. We can employ the dollar sign ($) before characters to convert a relative reference into an absolute reference.

4. True.

14

Charts in a Spreadsheet

One of the most popular features of OpenOffice Calc software is to generate charts based on numeric data. A chart is used to visualise data, making it easy to comprehend and analyse the data. Graphs are specifically designed to show changes and relationships in data, making them a powerful tool for data visualisation. Let us learn more about charts.

Charts

The term "chart" refers to graphic representation of data in which the data is visually represented through bars in a bar chart, lines in a line chart, or slices in a pie chart. A chart may present different information by representing tabular numerical data, functions, or certain types of quality structures.

Advantages of Using Charts

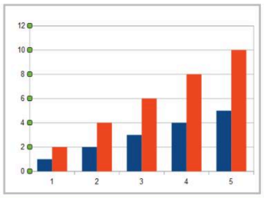
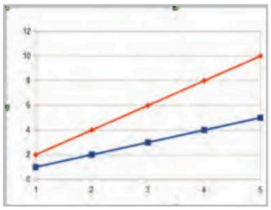
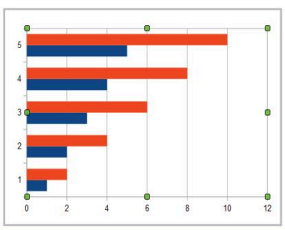
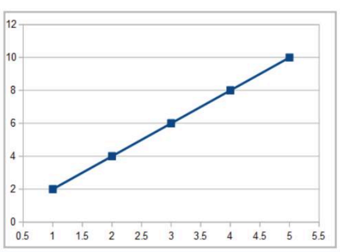
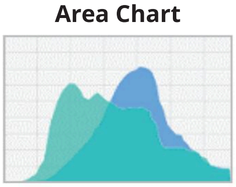
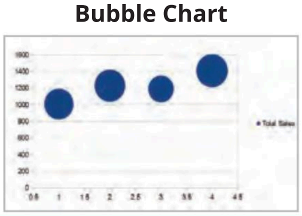
Here are the advantages of charts:

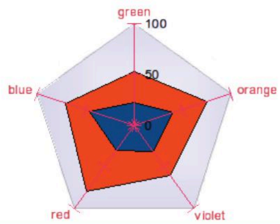
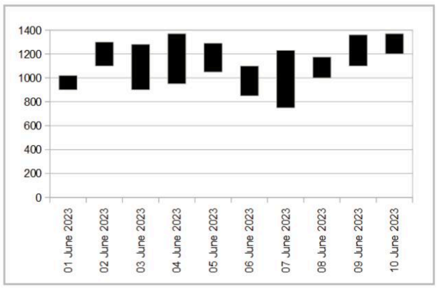
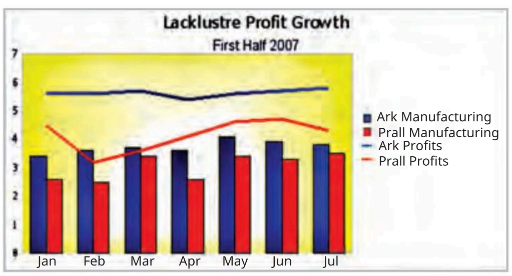
1. It improves the presentation and readability of the data.
2. It can be used to make certain judgments or do analysis.
3. It facilitates the concise and simple summarisation of a huge amount of data.
4. It facilitates more accurate data comparison.

OpenOffice Calc provides a variety of charts for analysing data. These charts have some characteristics in common that help users rationalise the data.

Types of Charts

Following are the different types of charts used in Calc.

Name of the Chart	Description
Column Chart 	This is the default chart type, as it is one of the most useful and easiest-to-understand charts. The x-axis shows categories. The y-axis shows the value for each category. It is used for displaying data that shows different trends over time.
Line Chart 	A line chart shows values as points on the y-axis. The x-axis shows categories. The y-values of each data series can be connected by a line. It is used to display quantitative values over a specified time interval.
Pie Chart 	A pie chart shows values as circular sectors of the total circle. The length of the arc, or the area of each sector, is proportional to its value. It is used to show data as a percentage of a whole.
Bar Chart 	This type of chart is used to represent data using horizontal bars. The length of each bar is proportional to its value. The y-axis shows categories. The x-axis shows the value for each category. It is used to summarise and compare values in a data category and provide a snapshot of your data at specified points in time.
Scatter or XY Charts 	An X–Y chart in its basic form is based on one data series consisting of a name, a list of x values, and a list of y values. Each value pair (x|y) is shown as a point in a coordinate system. It is used to observe and visually display the relationship between variables.
Area Chart 	An area chart is a graph that combines a line chart and a bar chart to show changes in quantities over time. The data points are plotted and connected by line segments and the area below the line is coloured or shaded. It is useful where you wish to emphasise the volume of change.
Bubble Chart 	A bubble chart can display data in three dimensions, which is used specifically for showcasing relationships in social, economic, medical, and other aspects. The bubble chart replaces data points with bubbles, and the size of the bubble represents the data value. It is often used to present financial data.

Name of the Chart	Description
Net Chart 	A net chart is useful for comparing data that are not time series but show different circumstances, such as variables in a scientific experiment. The poles of the net chart are equivalent to the y-axes of other charts.
Stock Chart 	A stock chart is a specialised column graph specifically designed to represent the price movement of stocks and shares. The data required for these charts is quite specialised, with series for opening price, closing price, and high and low prices. The x-axis represents a time series.
Column and Line Chart 	A column and line chart is a dual-axis chart that combines a column and a line chart. The two charts share an X-axis, but each has its own Y-axis. It is useful for combining two distinct but related data series, for example: sales over time (column) and profit margin trends (line).

Components of a Chart

Following are the different types of chart components used in Calc.

1. **Chart Title:** The Chart Title is used to display the name or aim of the chart.
2. **Chart Wall:** The chart wall is the "vertical" background behind the data area of the chart. It is a window within the chart area that includes the plotted data series, category, and value access in addition to the actual chart itself.
3. **Data Series:** Data series are the bars, slices, or other elements that show the data values. If there are multiple data series in the chart, each will have a different colour or style.
4. **Legend:** For clarity, legends provide a brief visual depiction of each data series in the chart. If the data contains numerous coloured visuals, legends will explain what each labelled visual signifies.
5. **Chart Area:** This includes all the areas and objects in the chart.
6. **Category Axis:** Category axis, or X-axis, is the horizontal axis.
7. **Value Axis:** The value axis, or Y-axis, is the vertical axis used to plot the values.
8. **Category Name:** Category names are the labels, which are displayed on the X and Y-axes.
9. **Gridlines:** These can either be horizontal or vertical lines, depending on the selected chart type. They extend across the plot area of the chart. Gridlines make it easier to read and understand the values.

Refer Fig. 14.1 to have a look at the components of the chart.

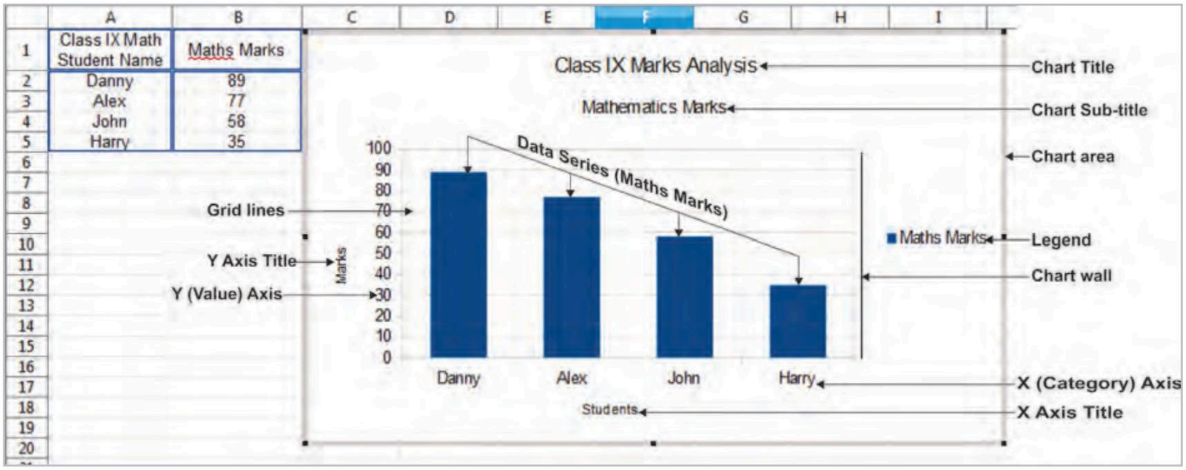


Fig. 14.1: Components of Chart

Creating a Chart

To demonstrate the process of making charts and graphs in Calc, let us consider the table below.

	A	B
1	**Class IX Maths Marks Analysis**	
2	**Student Name**	**Maths**
3	Danny	89
4	Alex	77
5	John	58
6	Harry	35
7		

Fig. 14.2: Marks Analysis

1. Select the cells that contain the data that you want to present in your chart. In the above example, select A3:B6.

2. Click on the **Insert** > **Chart** option or click the **Chart** icon on the Standard toolbar.

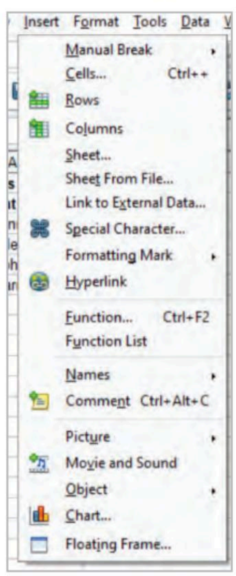


Fig. 14.3: Pop up Menu Having Chart Option

3. The **Chart Wizard** dialog box opens as shown in Fig. 14.4.

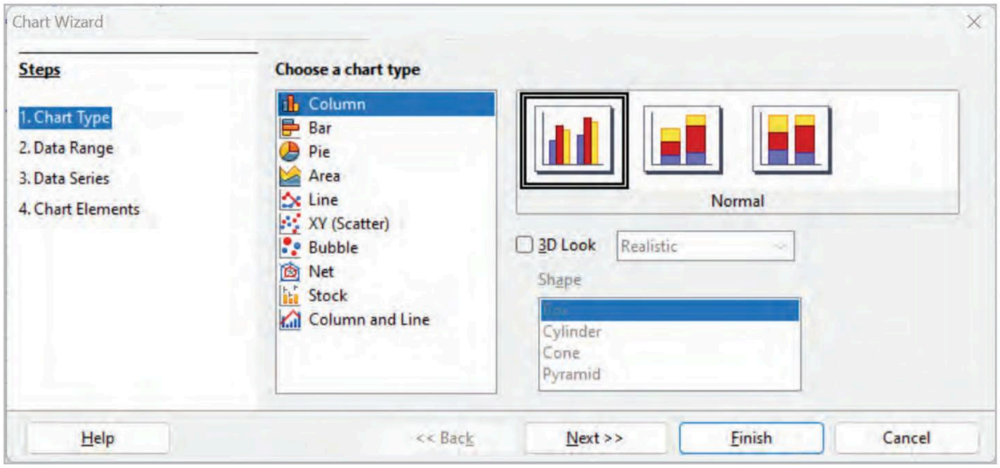


Fig. 14.4: Chart Wizard Dialog Box

4. The Chart Wizard has three main parts:
   - A list of the steps to insert the chart.
   - Choose a chart type.
   - The available options for each chart type.

   You can always go back to a prior step and make different choices by clicking on the **Back** button.
5. Choose a Chart type and its sub-type. Select **Column** as the chart type and then the **Normal** option. (Refer to Fig. 14.4). Then, click on the **Next** button.
6. In Step 2, **Data Range** is displayed according to the cells selected. Refer to Fig. 14.5.

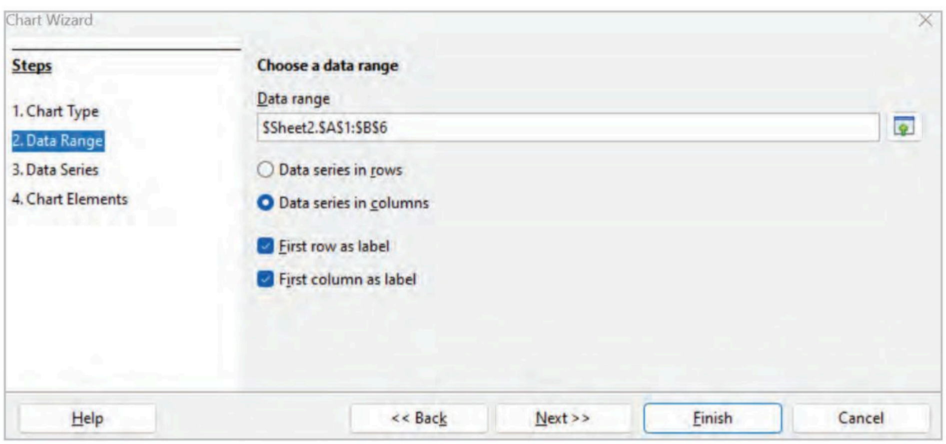


Fig. 14.5: Data Range

7. Select either the rows or columns option for the data series. Verify whether the data range has labels in the first row, the first column, or both. Then click on the **Finish** button or **Next** to make further changes to the chart's details.
8. In Fig. 14.5, click on **Data series in columns**. Check the **First row as label** and the **First column as label**. Click on **Next**.
9. In Step 3, in the **Data Series** list box, a list of all data series in the current chart is displayed. In the given example, only one data series (Maths) is displayed. Refer to Fig. 14.6.

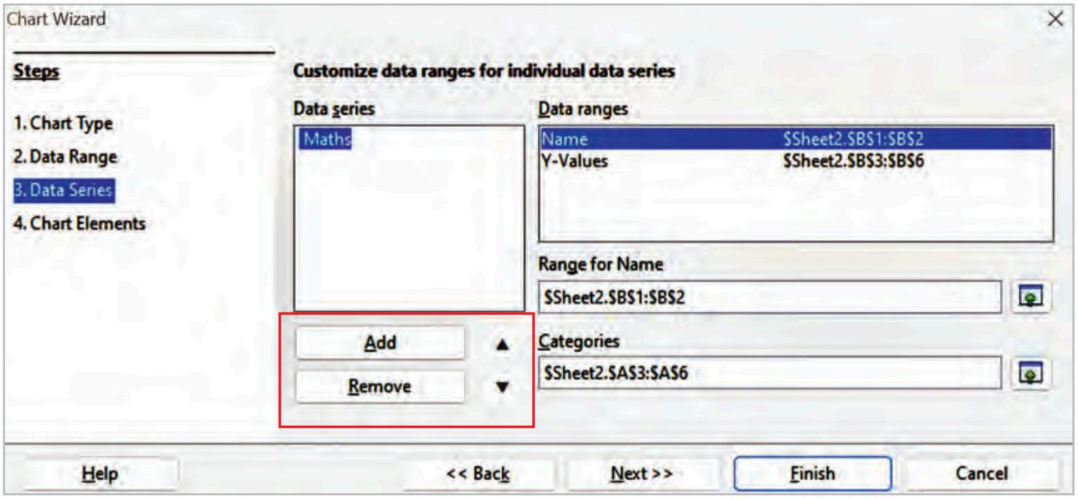


Fig. 14.6: Data Range for Data Series

10. To add a new data series below the currently chosen entry, click **Add**. The selected entry's type matches that of the new data series.

- To remove the chosen entry from the Data Series list, click on **Remove**.
- To move the currently chosen entry in the list up or down, use the **Up** and **Down** arrow buttons.

Then, select the **Next** button.

11. In Step 4, Chart Elements window is displayed as shown in Fig. 14.7.

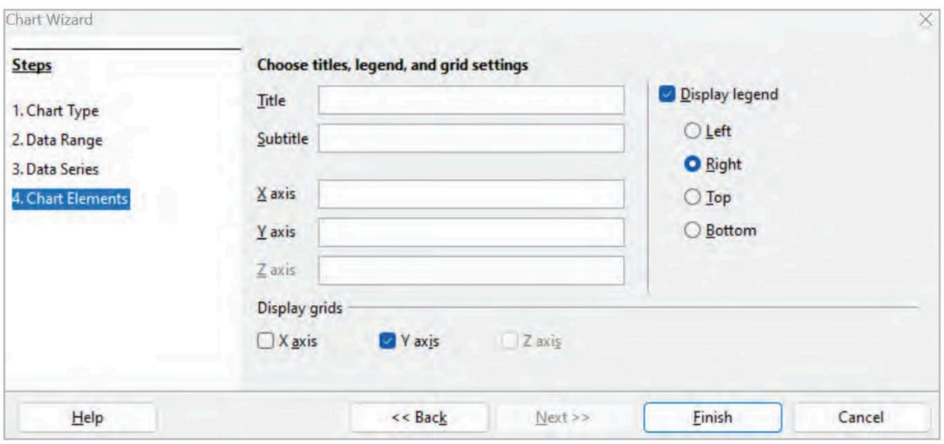


Fig. 14.7: Chart Elements

12. In the **Title** box, type "Class IX Marks Analysis".

In the **Subtitle** box, type "Mathematics Marks".

In the **X-axis**, type "Students Name".

In the **Y-axis**, type "Maths Marks".

In the **Display legend** check box, choose the **Right** option (default).

In the **Display grids**, check **Y axis**.

Fill in the above details as shown in Fig. 14.8.

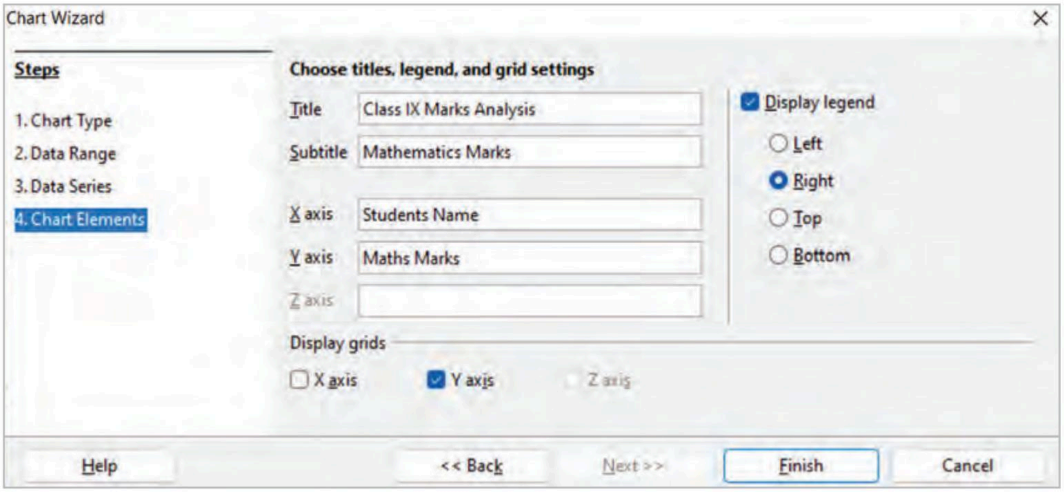


Fig. 14.8: Filled Values of Chart Elements

13. Then, click the **Finish** button. A Column chart (Refer to Fig. 14.9) is created as shown below.

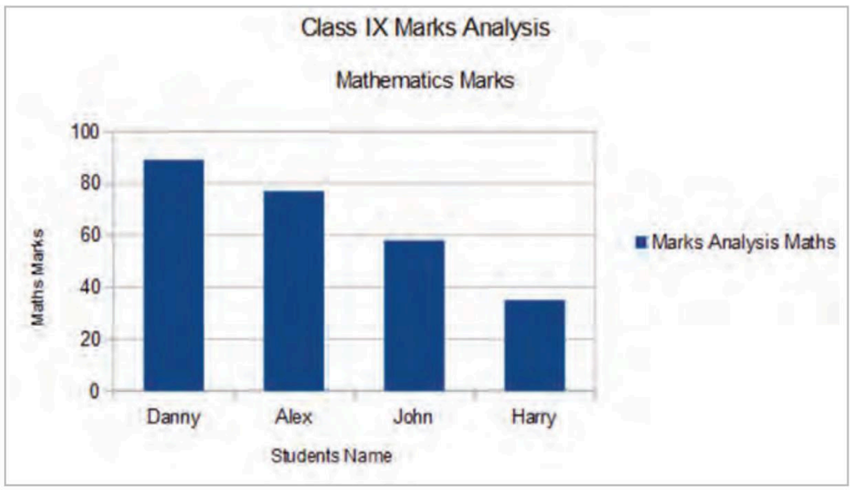


Fig. 14.9: Column Chart displaying Mathematics Marks Analysis

In a similar manner, the above data can be analysed with the help of different chart types, such as Bar chart, Pie chart, Line Chart, and so on.

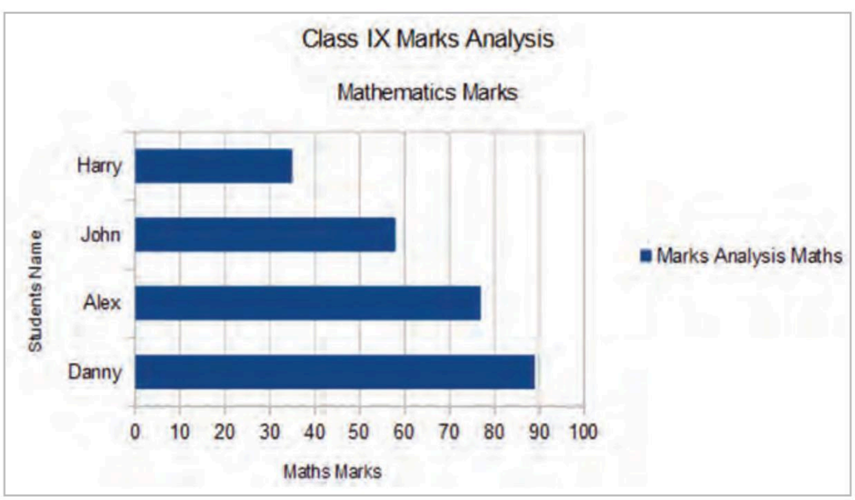


Fig. 14.10: Bar Chart Displaying Mathematics Marks Analysis

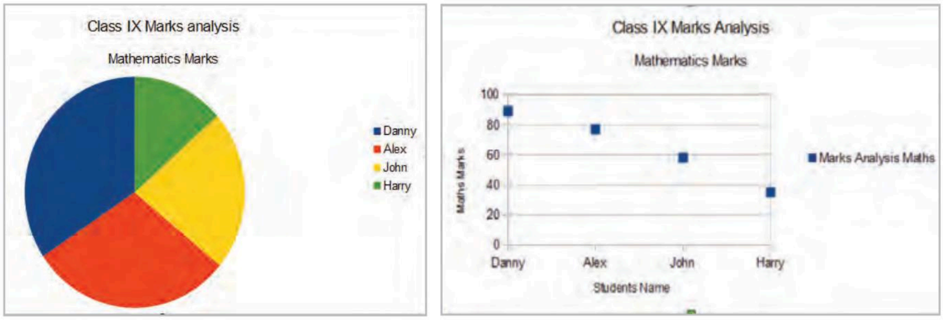


Fig. 14.11: Pie and Line Charts Displaying Mathematics Marks Analysis

Did You Know?

After you have created a chart, you may find things you would like to change. Through the Insert and Format menus, or by right-clicking on the chart and using the options in the Context menu, you may change the chart type, chart elements, data ranges, typefaces, colours, and many other parameters.

Activity Time

Activity 1: Input the following data into a spreadsheet and generate a 3D Column Chart to visualise the information. Also, give a suitable chart title. ***(Individual Activity)***

	A	B	C
1	**Name**	**Science Marks**	**Maths Marks**
2	Harry	13	12
3	Danny	98	45
4	Alex	72	76
5	John	68	87
6	Jerry	90	56
7			
8			
9			
10			

Activity 2: Input the following data into a spreadsheet and generate a Pie chart to visualise the information. ***(Individual Activity)***

Favourite Cheese Data	
Cheese Type	**Number of people**
Brie	9
cheddar	23
Dairylea	7
Lancashier	9
other	8
Red Leicester	7
Stilton	14
Wensleydale	12

Chapter Checkup

A **Select the correct option.**

1. Which chart will show the direction of change in numbers over a period of time by connecting data points?
   - **a** Bar **b** Line **c** Pie **d** Column
2. Bar Charts are most effectively used to:
   - **a** Present scientific data **b** Present stock data
   - **c** Show parts of a whole **d** Compare groups of data
3. .... is a pictorial representation of worksheet data.
   - **a** Flowchart **b** Chart **c** Picture **d** Graphic

B **Fill in the blanks with the most suitable words.**

1. .... is the default chart type in Calc.
2. .... are the bars, slices, or other elements that show the data values.
3. A .... is the key to a chart; it explains the significance of each labelled visual in a chart.
4. A .... compares pieces of data using horizontal stacks.

C **State whether the following is *True* or *False*. Correct the statements that are false.**

1. Graphs are not good at showing changes and relationships in data.
2. The length of the arc, or the area of each sector, is proportional to the data value in a pie chart.
3. Graphs make it hard to identify the trends in the data.
4. The value axis, or Y-axis, is the vertical axis used to plot the values.

D **Answer the following questions. *(Solved)***

Q1. What is a chart wall?

A1. The chart wall is the "vertical" background behind the data area of the chart. It is a window within the chart area that includes the plotted data series, category, and value access in addition to the actual chart itself.

Q2. Differentiate between a column chart and a bar chart.

A2. a. A bar chart is oriented horizontally, whereas a column chart is oriented vertically.

b. Bar charts are suitable for displaying long data labels. Column charts have limited space on the category axis. So, when your data labels are long, the category axis may look cluttered.

Q3. Shally has collected data on how much each member of her class spends on magazines each month. She wishes to display this information in a graph.

a. Name one suitable graph that Shally could use.

b. Explain how Shally could use the OpenOffice Calc to produce the graph.

A3. a. A bar graph is an appropriate graph that Shally may use to represent this information. Each student spending on magazines can be represented by a separate bar, making it easy to compare different spendings by different individuals.

b. Shally can enter student names and expenditure information into OpenOffice Calc.

Then, she can choose the data, select a bar chart type, add appropriate chart title, add category names, save the graph, and integrate it into her project.

Answer Key

A **1.** b **2.** d **3.** b

B **1.** Column Chart **2.** Data series **3.** Legend **4.** Bar Chart

C **1.** False. Graphs are specifically designed to show changes and relationships in data, making them a powerful tool for data visualisation.

2. True.

3. False. Graphs generally make it easier, not harder, to identify the main point or trends in the data.

4. True.

Unit Reflection

Key Terms

Spreadsheet: A spreadsheet or worksheet is a file that is used for storing, organising, and manipulating data in a tabular format.

AutoFill: The AutoFill feature allows you to quickly fill cells with repetitive or sequential data, such as chronological dates, numbers, or repeated text.

Functions: Functions are in-built formulas that are used for doing calculations using the data entered.

Labels: A text or special character, descriptive information (non-numeric data), or a combination of numeric and alphanumeric data is treated as a label for rows or columns.

Concatenation: Joining two or more strings of text data is called concatenation.

Mathematical Operators: Symbols of calculations such as '+', '–', '*', '/', and '^' are called operators.

Arguments: Arguments are the values that are input to functions. These values can be in the form of text, numbers, logical values (True or False), constants, range of cells, or some other functions.

Data Formatting: Data formatting is the process of changing the appearance and layout of a cell or range of cells to make them easier to read and more visually pleasing.

Text Wrapping: Text wrapping allows you to rapidly change the dimensions of a cell so that all the text neatly fits inside it rather than extending outside the boundaries of the cell.

Cell Referencing: Cell referencing refers to a cell or range of cells on a spreadsheet and can be used in a formula, so that the values or data associated with that cell can be utilised for calculation in formulas.

Chart: A chart refers to a graphic representation of data in which the data is visually represented through bars in a bar chart, lines in a line chart, or slices in a pie chart.

Legends: Legends provide a brief visual depiction of each data series in the chart.

Things to Remember

- The AutoSum feature helps add the numeric values in a group of selected cells.
- OpenOffice Calc is the spreadsheet application of the Apache OpenOffice suite. This is a free-to-use, open-source program that can be used separately on multiple operating systems.
- The data that is to be moved needs to be cut from the original source, whereas the data that needs to be duplicated, needs to be copied from the original source.
- On applying a formula to a cell, the value of the equation is displayed in the cell, and the formula is displayed in the formula bar.
- There are four types of horizontal alignments: left, centre, right, justified, and filled.

- The orientation of the cell is a significant formatting characteristic that describes the direction in which the contents of the cell are to be displayed or printed.
- The orientation of the cell is a significant formatting characteristic that describes the direction in which the contents of the cell are to be displayed or printed.
- Only text can be used in selection lists. The selection list suggests text that has already been typed in the same column.
- Calc provides three major types of cell references: relative cell reference, absolute cell reference, and mixed cell reference.
- The dollar sign fixes the reference in such a way that it remains unchanged even when you copy or apply a formula to other cells.
- Data series are the bars, slices, or other elements that show the data values.
- Gridlines can either be horizontal or vertical lines, depending on the selected chart type. They extend across the plot area of the chart. Gridlines make it easier to read and understand the values.

Test Your Knowledge

A. Select the correct option.

1. The ribbon present at the bottom of worksheets that shows the current information on the worksheet is the .... bar.
   - a. Status
   - b. Horizontal scroll bar
   - c. Formatting toolbar
   - d. Standard toolbar
2. The .... formula use more than one operator in a single calculation.
   - a. Customised
   - b. Basic
   - c. Text
   - d. Compound
3. .... in text wrapping is a feature that automatically adds hyphens to break words at the end of lines when text is being wrapped.
   - a. Hyphenation
   - b. Shrinking
   - c. AutoFill
   - d. Alignment
4. A .... cell reference consists of an absolute column and a relative row, or a relative column and an absolute row.
   - a. Absolute
   - b. Mixed
   - c. Relative
   - d. Hybrid
5. A .... chart can display data in three dimensions, which is used specifically for highlighting relationships in social, economic, medical, and other aspects.
   - a. Net
   - b. Bar
   - c. Bubble
   - d. Pie

B. Fill in the blanks with the most suitable words.

1. New, Open, Save As, Page Preview, Print, etc., are the commonly used commands in the .... menu.
2. When an .... cell is clicked, it gets highlighted by a thick black border.
3. The .... function gives us the smallest of the entered values among the selected cells.
4. If you want to display the numerical values with a leading zero, then this type of data can be formatted as ....
5. The .... axis or X-axis is the horizontal axis of a chart.

C. State whether the following is *True* or *False*. Correct the statements that are false.

1. Google Sheets is a type of spreadsheet software application.
2. Ctrl+C is the shortcut key to paste data.
3. The Date format is used in spreadsheets to consider your input as a calendar date instead of a regular number.
4. "=B2:D6" without the quotation marks refers to the entire cell range from cell B2 to cell D6.
5. A chart can be used to do analysis.

D. Short answer-type questions.

1. What is a spreadsheet?
2. What is an argument?
3. What is relative cell referencing?

E. Long answer-type questions.

1. What are the advantages of using a chart?
2. What is data formatting? Give examples.
3. What are the elements of a formula?

F. Competency-based questions.

1. Richa is computing her monthly expenses, for which she needs to add numbers from multiple columns. Name the button that can help her compute total expenses.
2. Utsav is using OpenOffice Calc software to enter marks in various subjects for all his classmates. As this is huge data, it has now become difficult for him to analyse the data. What feature of Calc must he use for better data visualisation and analysis?